RESTORATIVE DENTISTRY

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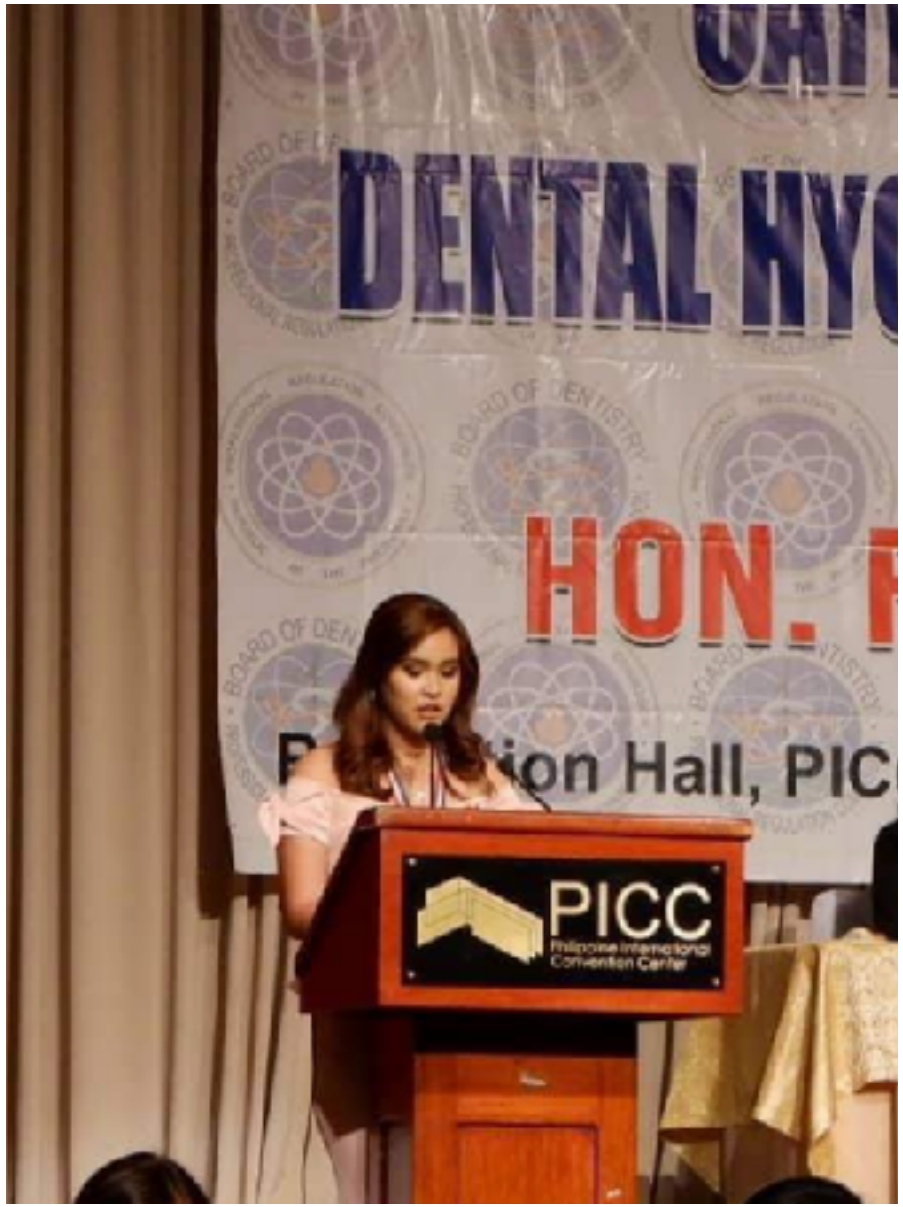
Cum Laude, Centro Escolar University-Manila

Top 1-Dentist Licensure Examination 2018

Master of Science in Dentistry Major in Prosthodontics (University of the East)



AWARDING OF CERTIFICATE

University of the East
MANILA

Mandaluyong

Pasig

Pampanga

Quezon City

DMF DENTAL TRAINING CENTER BY: DR. C. GUERRERO-PINEDA

1. Which structure of the tooth doesn't regenerate?

A. Enamel
B. Dentin
C. Cementum

Regeneration ≠ Remineralization

2. Which structure of the tooth resembles bone chemically and physically?

A. Enamel

B. Dentin

C. Cementum

COMPOSITION OF TOOTH STRUCTURE

	INORGANIC	ORGANIC
ENAMEL (substantia adamantina)	96-98 % calcium and phosphate ions (hydroxyapatite crystal)	2-4 % Enamelin proteins (amelogenins)
DENTIN (substantia eburnea)	70 % (same as bone) calcium and phosphate ions (smaller hydroxyapatite crystals than enamel)	30 % 20 % Collagen fibre mesh (type I, III, *IV*, V, VI) 10% water
CEMENTUM (substantia ossea)	50 % Hydroxyapatite crystals	50% Matrix proteins (collagen type I, *III, XII*) 12% water

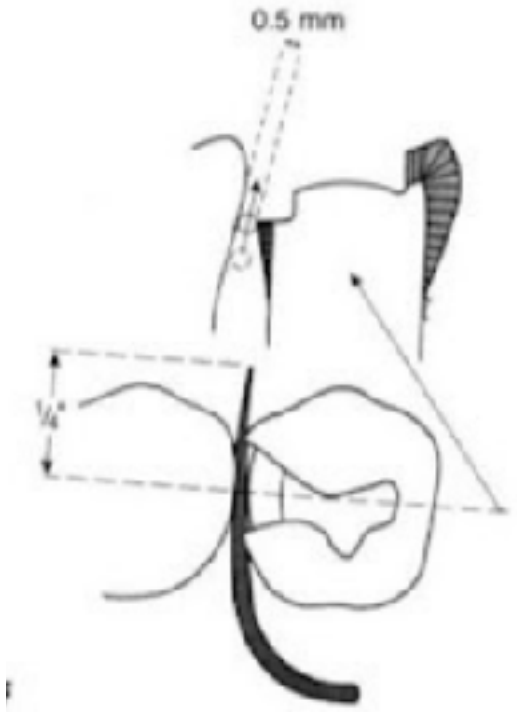
3. In class II cavity preparation, the cavosurface margin of the gingival seat clears the adjacent tooth by

A.0.20 ± 0.05mm

B.0.50 ± 0.20mm

C.0.80 ± 0.35mm

D.1.10 ± 0.45mm



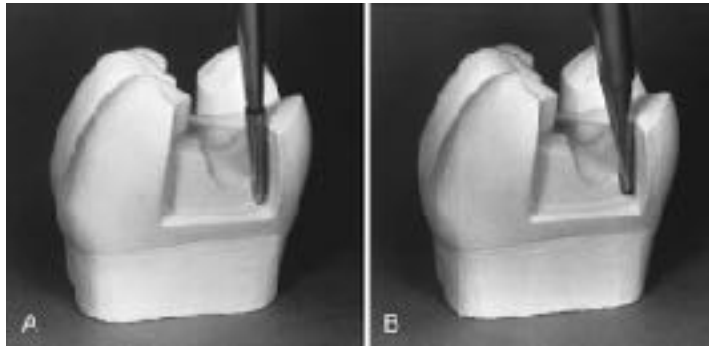
4. Retentive grooves in the proximal box of a class II amalgam cavity should be

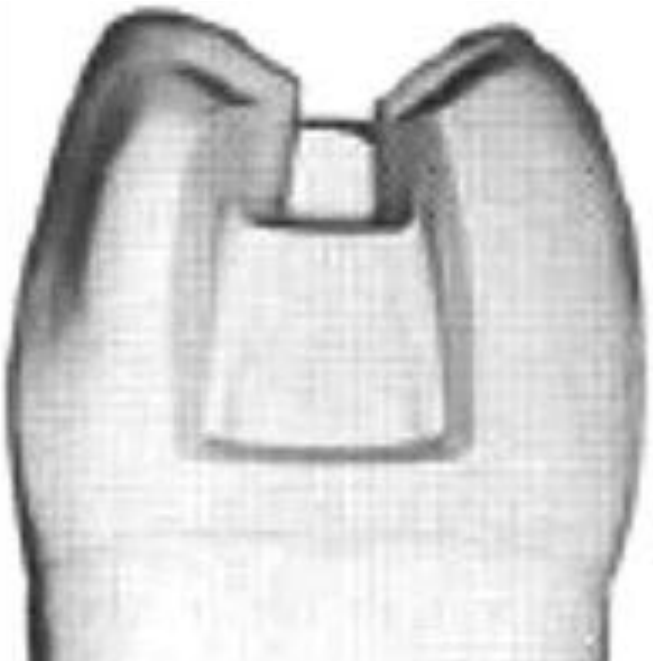
A. Sharp and elongated at the dentinoenamel junction

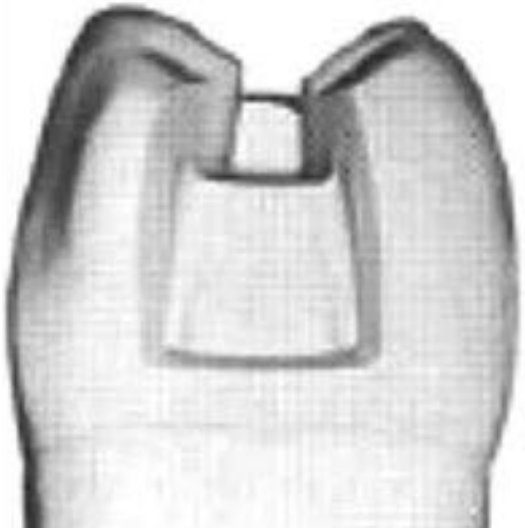
B. Round and elongated in the dentin

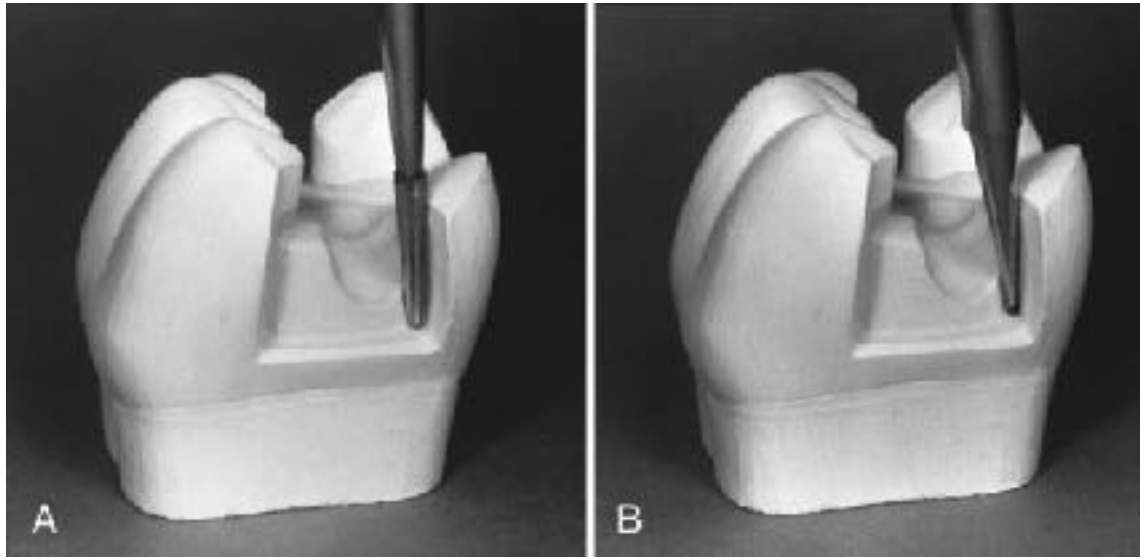
C. Round and short in the dentin

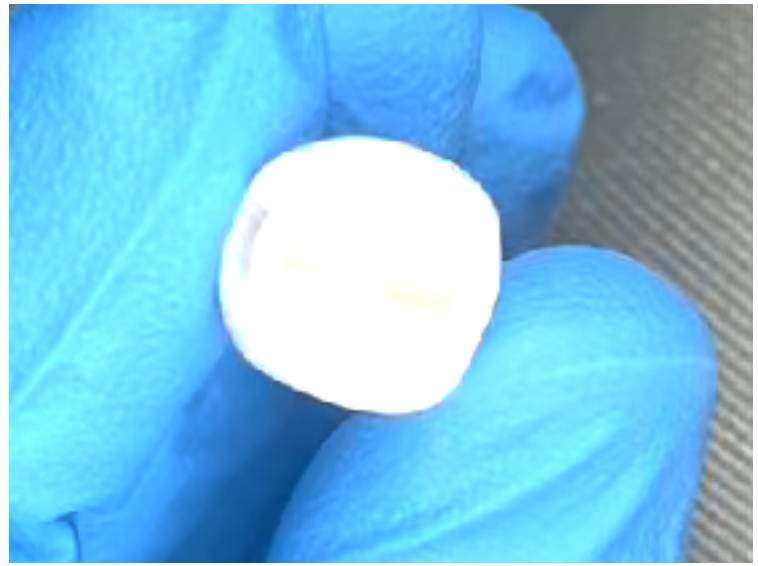
D. Sharp and elongated in the dentin











✓ ***<u>Retention grooves/locks:</u>***

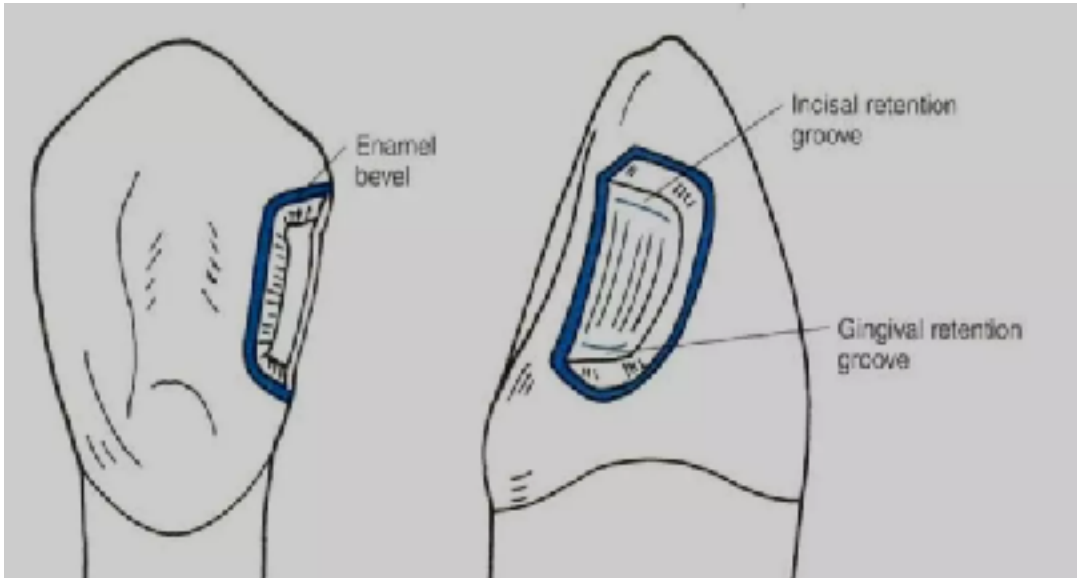
-prepared with 169L bur or ¼ round bur

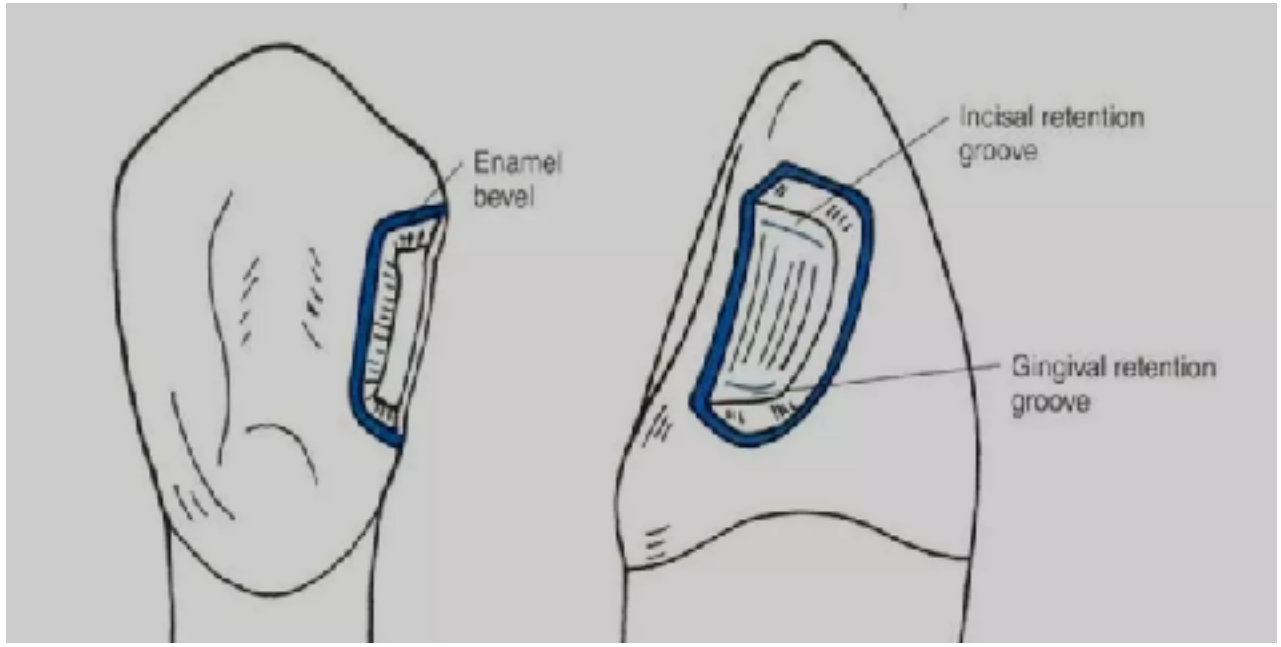
-prepared 0.5mm depth of translation of tip of 169L bur at gingival floor level

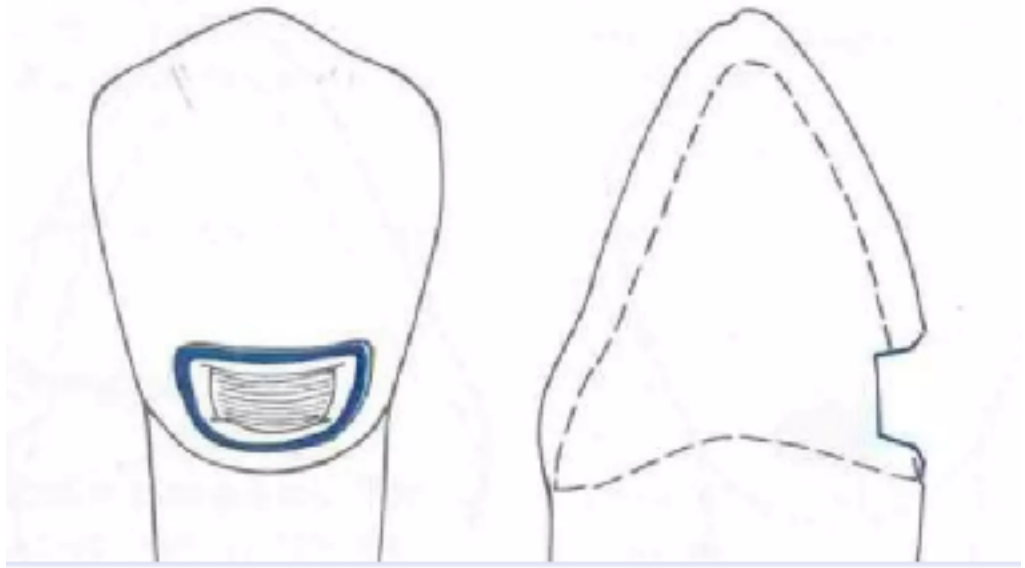
-0.2mm inside DEJ (not to undermine enamel)

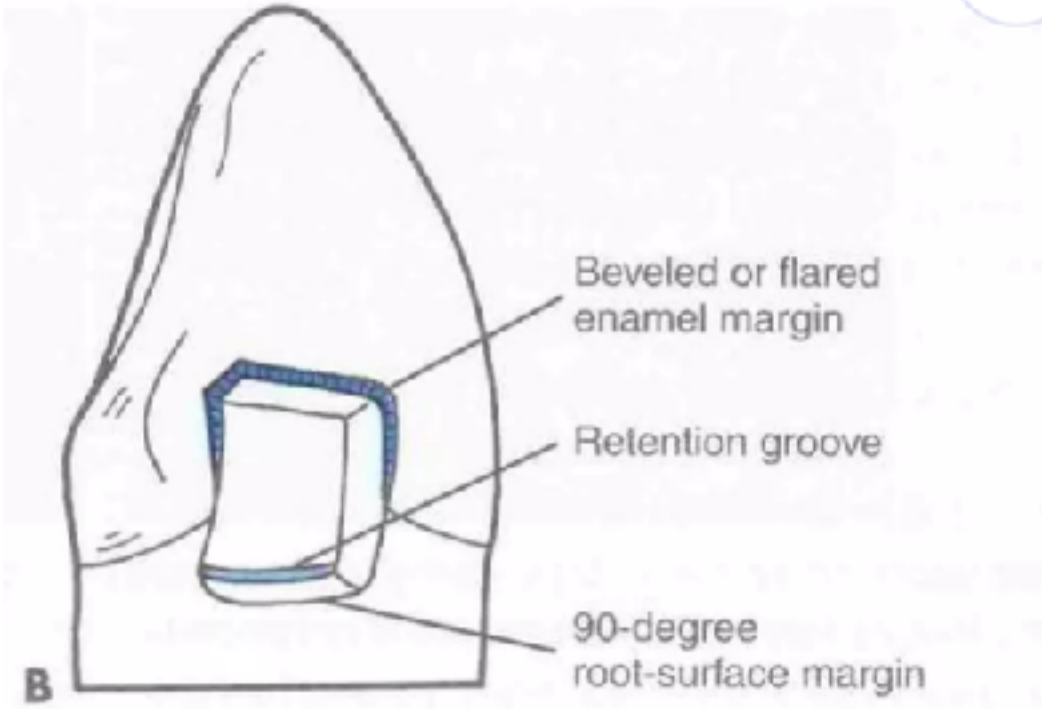
5.the retention for Class III restoration are placed:

A. In mesial and distal areas
B. Occlusal and gingival area
C. In all except the axial wall
D. At the expense of axial wall



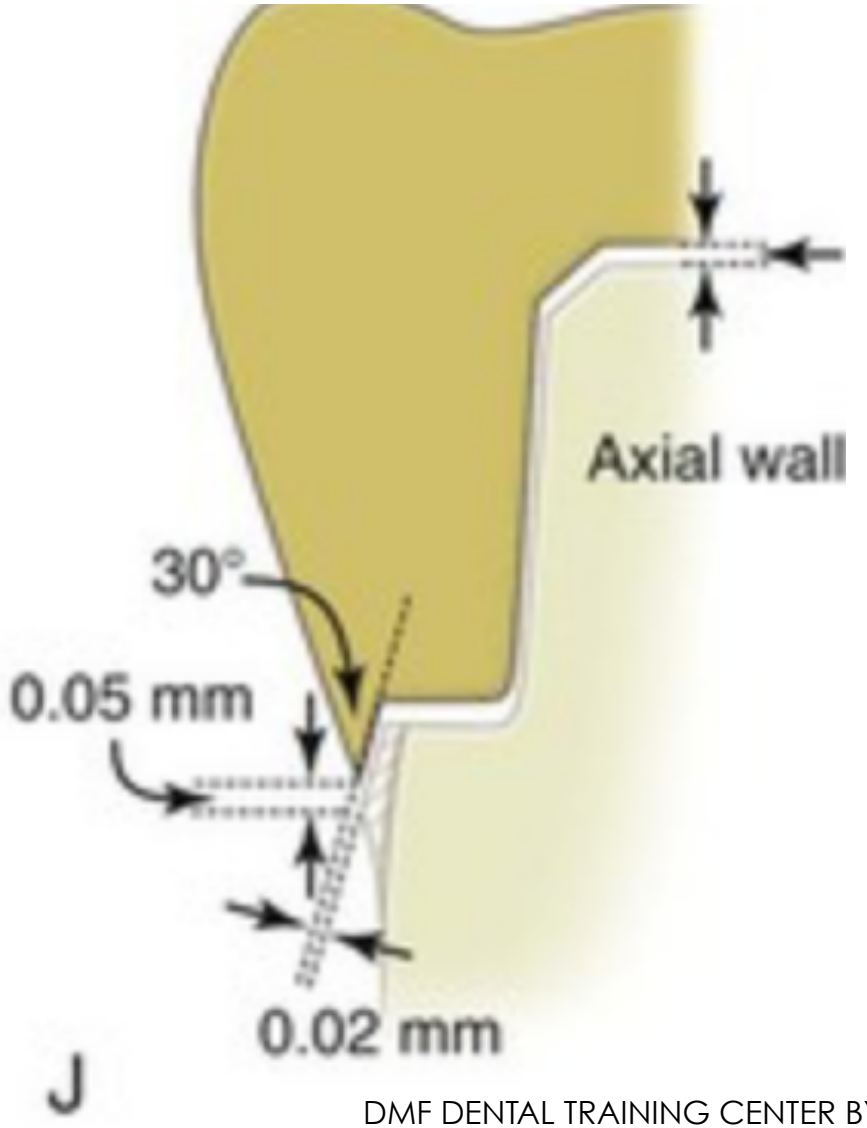

Enamel bevel
Incisal retention groove
Gingival retention groove



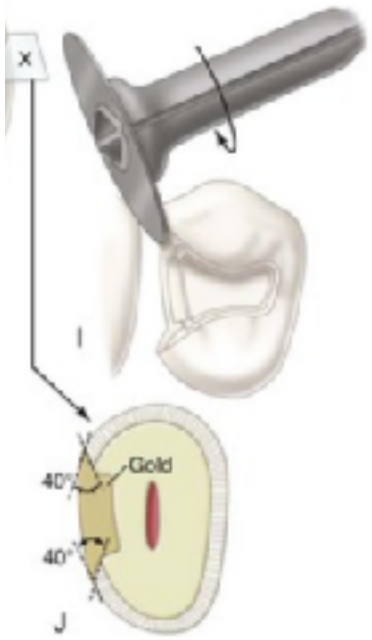

Beveled or flared enamel margin
Retention groove
90-degree root-surface margin
B

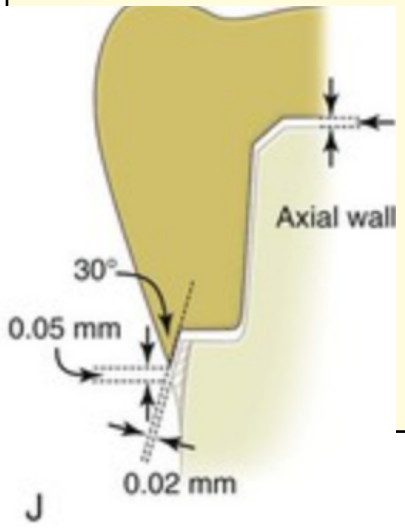
6. Gingival cavosurface margin bevel in a class II inlay preparation is

A. 15-20 degrees
B. 40 degrees
C. 30 degrees
D. 45 degrees



COMPARISON OF THE CAVOSURFACE MARGINS FOR A CONSERVATIVE CLASS II AMALGAM AND GOLD INLAY PREPARATION		
Characteristics	**Class II gold inlay**	**Class II amalgam**
Occlusal CSM	Beveled to result in 40-degree marginal metal ***(140-150 cavosurface angle)***	90 degree angle with the external surface
Proximal CSM	Extended up to the embrasures to prevent recurrent caries. Beveled to result in 40 degree marginal metal	90 degree angle with the external surface
Gingival CSM	Beveled to result in 30 degree marginal metal All 40 deg except Gingival CSM and Counterbevel of onlay	Beveled in permanent teeth (to produce 90-degree amalgam marginal angle (15-20 degrees) NO BEVEL IN DECIDUOUS!





7. For a patient with gingival recession, the best technique of wedging is

A. Piggy-back wedging — Deep proximal box, gum recession, or both

B. Double wedging — Proximal box is wide faciolingually

C. Wedge-wedging — Proximal concavity gingival to contact (MO upper 1st pm)

D. Single wedging

TYPES OF WEDGING

1. SINGLE WEDGING

TYPES OF WEDGING

2. PIGGY-BACK WEDGING

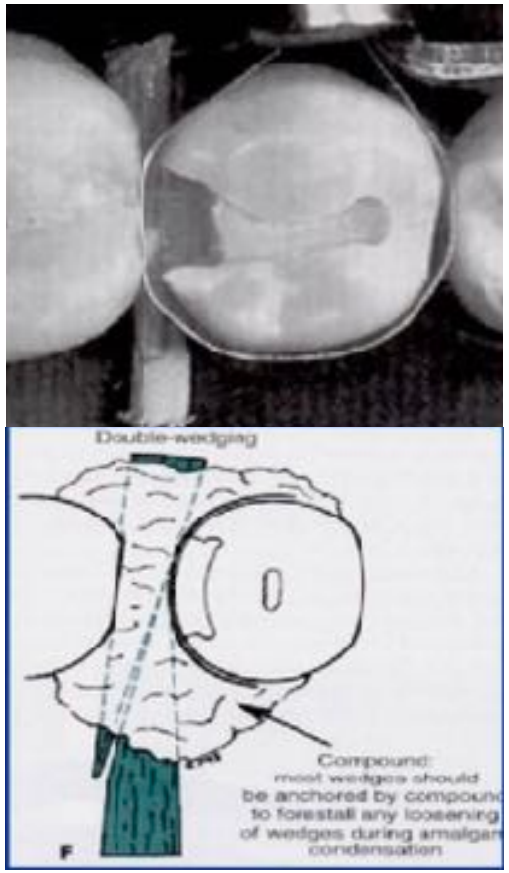
-wedge is significantly apical to the gingival margin, a smaller 2nd wedge is piggy-backed on the 1st

INDICATION: proximal box is DEEP gingivally or interproximal tissue has receded or both

TYPES OF WEDGING

3. DOUBLE WEDGING

-inserting 2 wedges, one from lingual & a 2nd wedge from facial embrasures

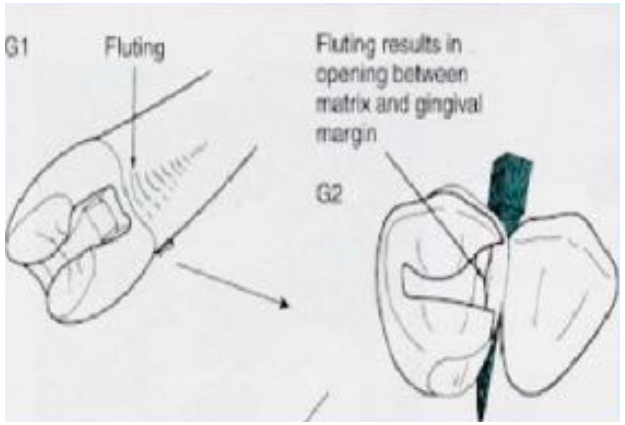


INDICATION:
faciolingually wide proximal boxes to provide maximal closure of band along gingival margin

TYPES OF WEDGING

4. WEDGE WEDGING

-a 2nd wedge is inserted from lingual or facial embrasure, between the 1st wedge & the band



INDICATION:
proximal concavity, gingival to the contact (mesial of maxillary 1st premolars)

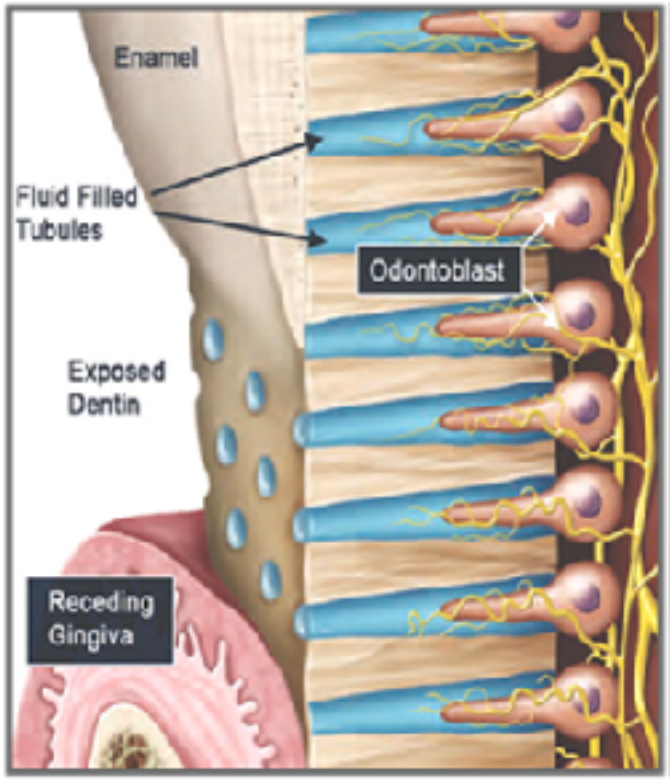
8. A stimulus on a tooth such as cold or air drying can cause fluid movement to become rapid, which is then interpreted as pain to the nerve endings in the pulp. This theory is known as the:

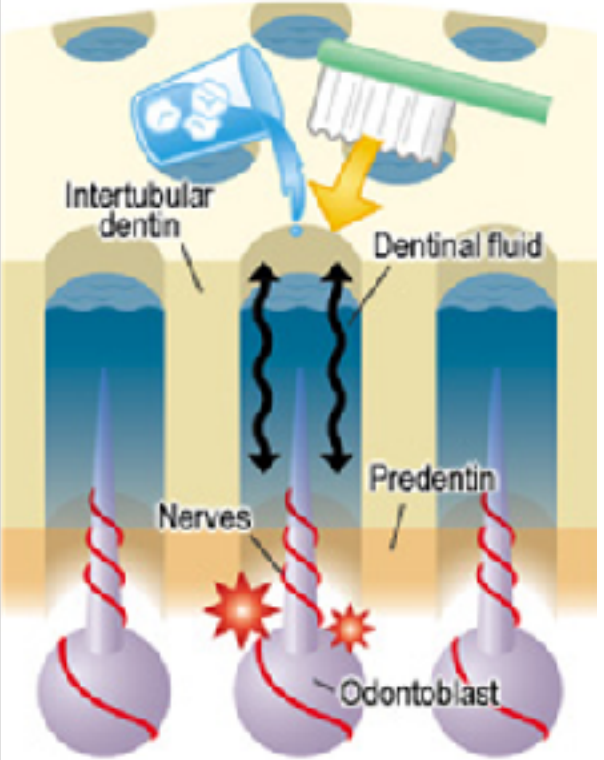
A. Pulpal inflammation theory

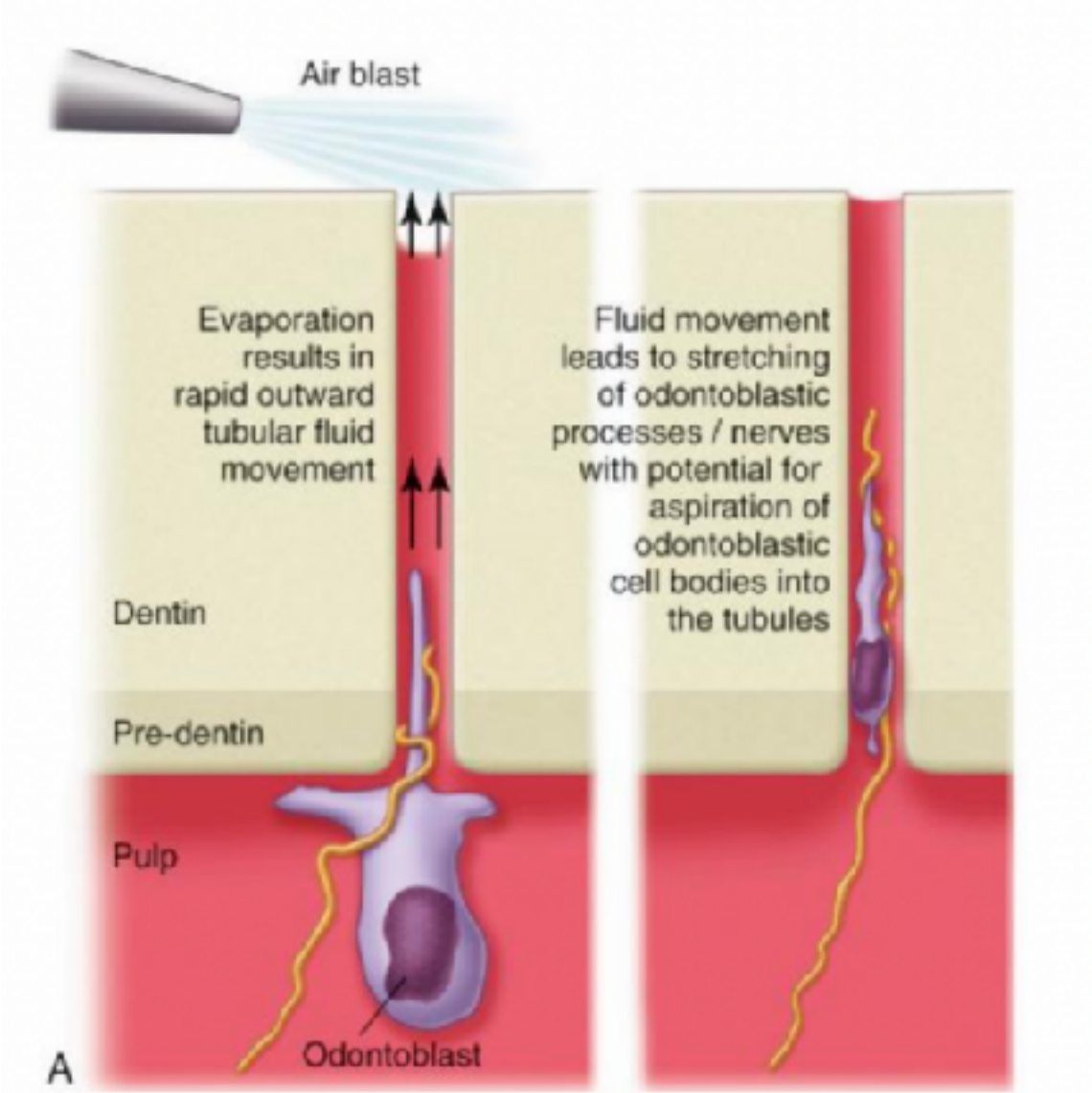
B. Hydrodynamic theory

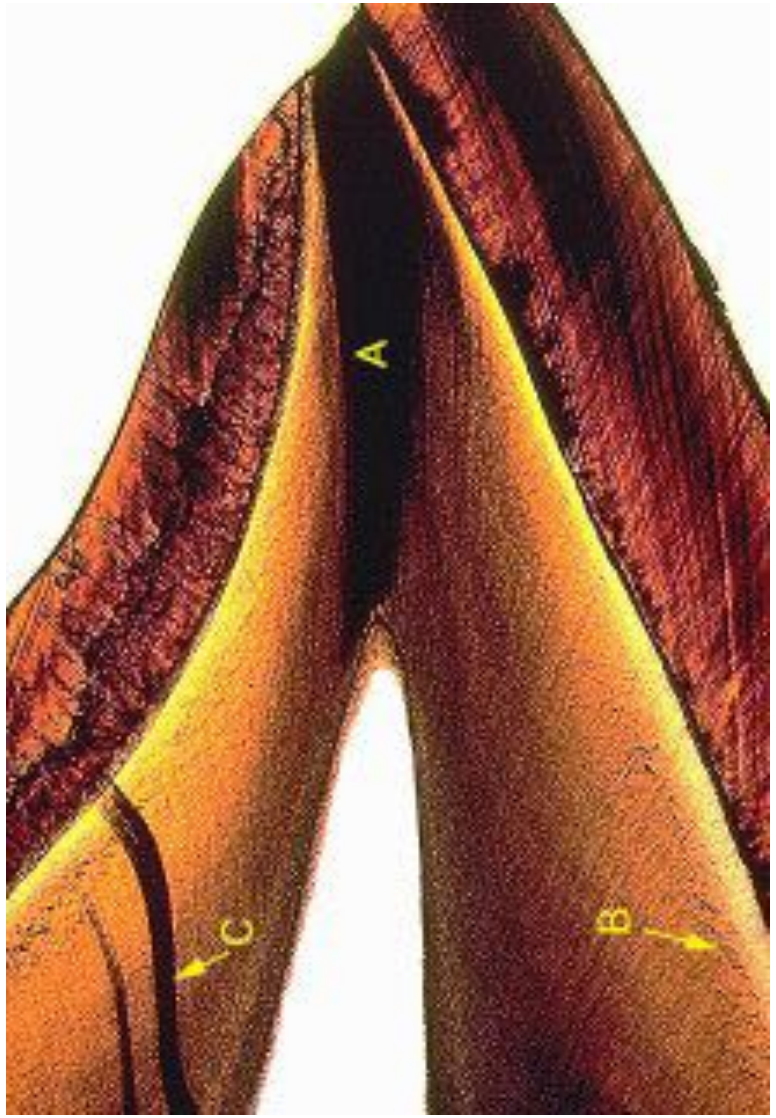
C. Thermal sensitivity theory

D. Thermal expansion theory



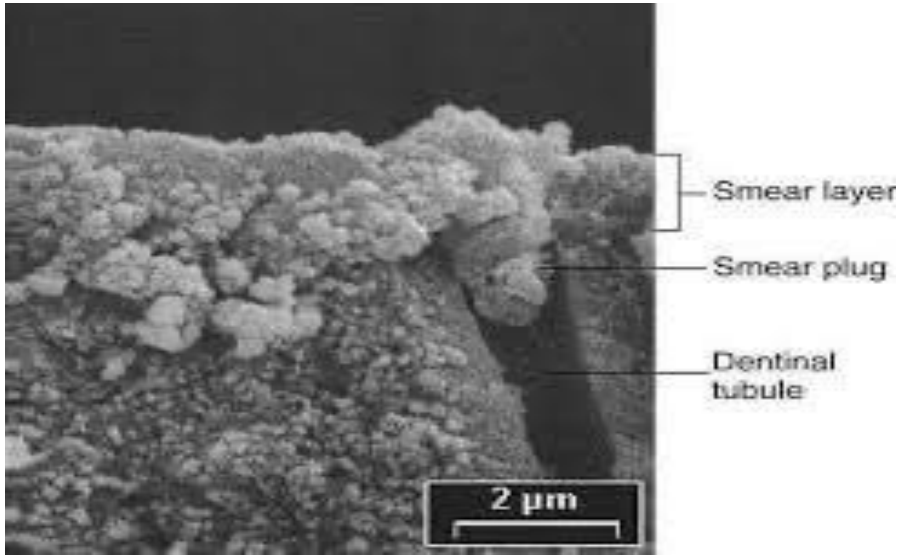


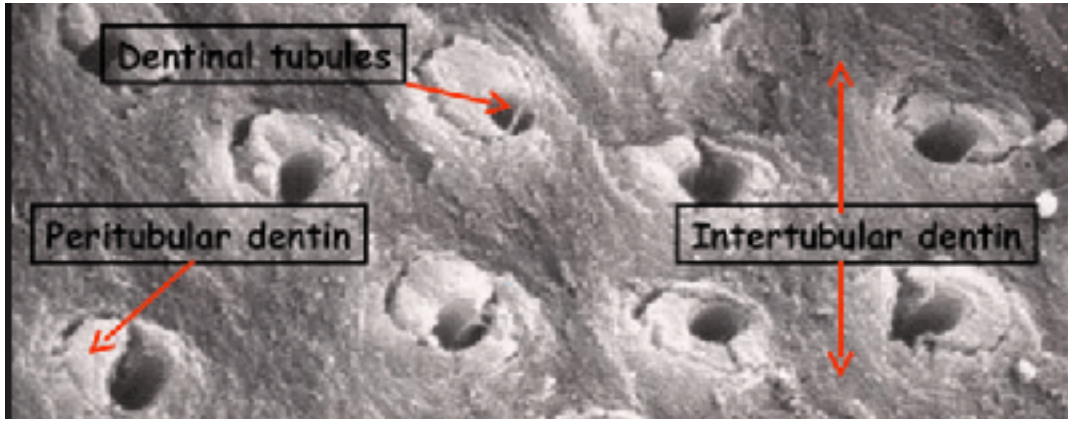

Air blast
Evaporation results in rapid outward tubular fluid movement
Fluid movement leads to stretching of odontoblastic processes / nerves with potential for aspiration of odontoblastic cell bodies into the tubules
Dentin
Pre-dentin
Pulp
Odontoblast
A


A
B
C

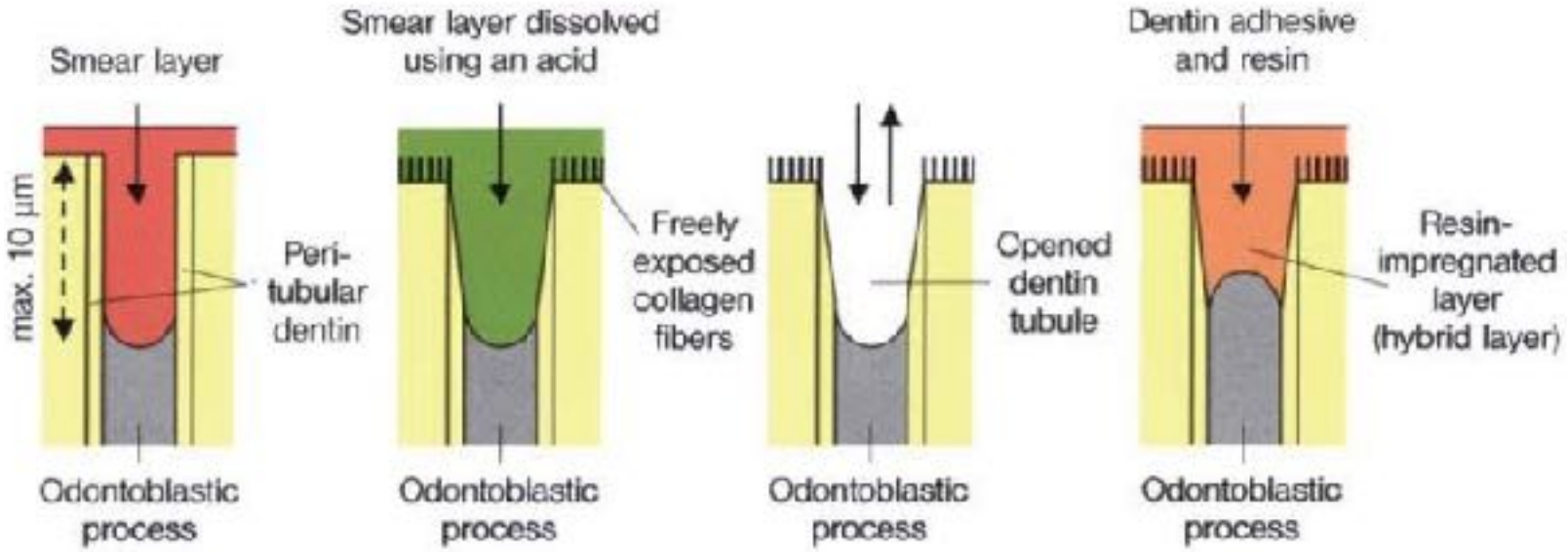
9. Which one of the following is the most important event for dentin bonding?

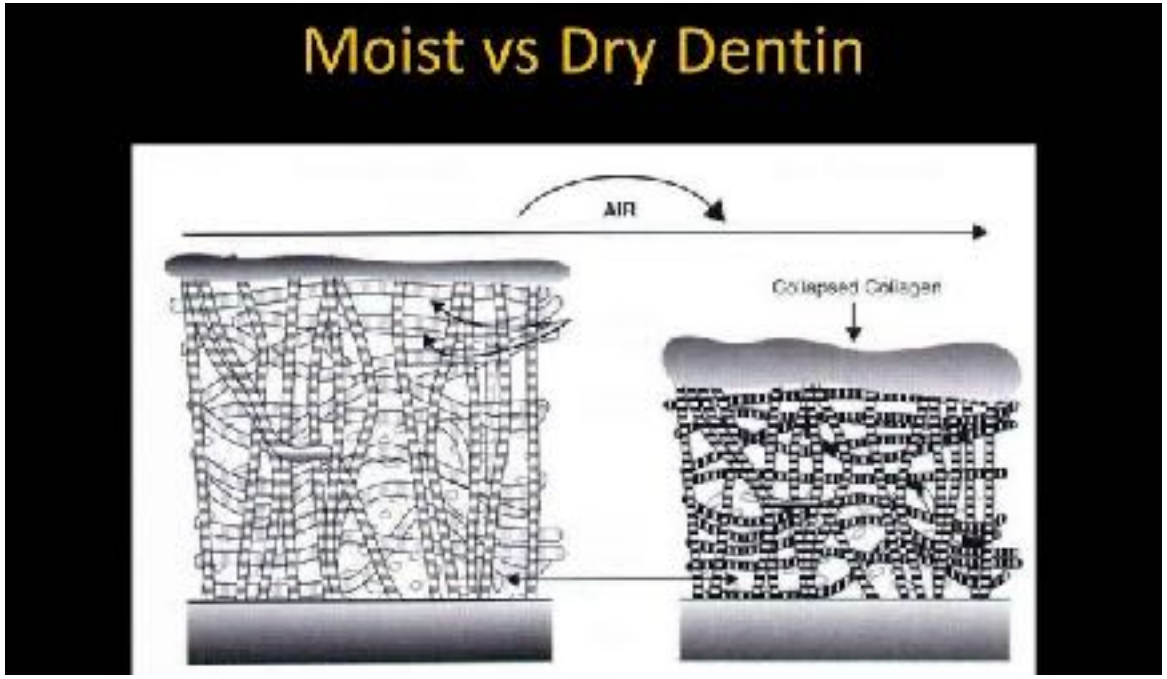
A. Smear layer removal
B. Smear plug removal
C. Peritubular dentin decalcification
D. Intertubular dentin decalcification





-collagen matrix (type I collagen)
-less mineralized than peritubular dentin





DO NOT OVERDRY:
collagen fibers will collapse
-adhesive cannot penetrate into the dentinal matrix

10. Kaolin in dental porcelain act as

A. Strengthener Quartz (fillers)

B. Binder Kaolin

C. Pigment

D. Crack minimizer Leucite, alumina (fillers)

COMPOSITION OF DENTAL CERAMICS/PORCELAIN

FELDSPAR (60-80%)	*an-hydrated potassium & sodium aluminum silicate *upon firing, they fuse, become glassy *acts as **matrix** for the quartz *gives the porcelain **translucency**
QUARTZ (15-25%)	*fine crystalline dispersion throughout vitreous phase *remains unchanged during firing *acts as **strengthening agent (filler)**
KAOLIN (3-5%)	*hydrated aluminum silicate (white claylike material) *becomes sticky during mixing, act as **BINDER** for the materials when the porcelain is still unfired *gives opacity to the mass **"OPAQUENESS"** *during firing, it adheres to quartz then shrinks

Feldspathic porcelain:

-Recent dental porcelain which does not contain KAOLIN
-more TRANSLUCENT

COMPOSITION OF DENTAL CERAMICS/PORCELAIN

GLASS MODIFIER (8-20%)	*alkali metal ion, sodium, potassium, calcium ***lowering the fusing temperature** of porcelain to be suitable for different applications *increasing **fluidity**
FLUX (9-15%)	*low fusing glass (boron oxide based glass) *action similar to glass modifier, **decreases sintering temperature**
SUGAR & STARCH	*help in forming workable mix instead of kaolin in feldspathic porcelain
PIGMENTS (1%)	*provides different & natural shades

11. To polish a restoration, you will likely use a bur with less cutting blades. This is because a bur with less cutting blades will cut more efficiently

More blades = less efficient tooth reduction = smoother cut

A. Both the statement and the reason are correct & related
B. Both the statement and the reason are correct but not related
C. The statement is correct, but the reason is not
D. The statement is not correct, but the reason is correct

12. Smooth finish is achieved by using

A.8 fluted bur
B.12 fluted bur
C.16 fluted bur
D.30 fluted bur

DENTAL BURS

-rotary cutting instruments that have bladed cutting heads

CUTTING BUR: average 6 blades
FINISHING BUR: average 12 blades

13. Type of glass ionomer used for ART (atraumatic restorative treatment)

A. Type I
B. Type III
C. Type V
D. Type VIII

Glass ionomers:

introduced by ***Wilson & Kent*** in early 1970s

Type I – Luting
Type II - Restorative
Type III - Base / Liner
Type IV – Pit and Fissure Sealants
Type V – Luting for Orthodontic purposes
Type VI – Core Buildup material
Type VII – High Fluoride releasing command set
Type VIII – (ART) Atraumatic restorative treatment
Type IX – Pediatric Glass Ionomer Cement

***high-viscosity glass ionomers**

14. The porcelain constituent that fuses at firing temperature and forms a matrix is

A. Kaolin — Acts as binder, and also gives OPACITY to the mass

B. Quartz — Strengthening agent (fillers)

C. Feldspar — Matrix, gives TRANSLUCENCY

D. Aluminum oxide

15. Dental Pulp under any restoration is best protected by

A.Cavity liner
B.Cavity base
C.2 mm thick dentin
D.All of the above

The single most important factor in protecting the pulp from trauma, thermal, or chemical change: ***DENTIN***

16. Instrument formula for ART

A. 13-7-14 *Or 15-7-14 (BIN-ANGLE SPOON EXCAVATOR)*

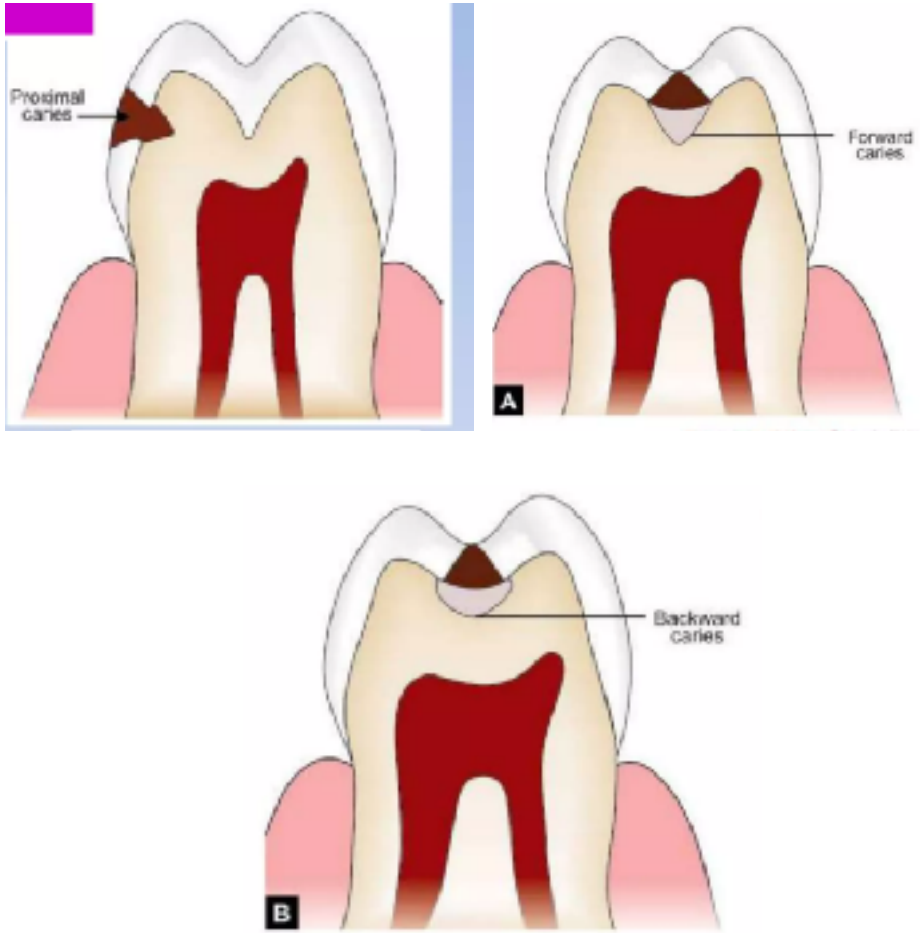
B. 10–6-14 *ENAMEL HATCHET*

C. 15-89-9-12 *GMT*

D. A & B

17. Whenever the caries cone in enamel is larger or at least the same size as that in dentin, it is called as

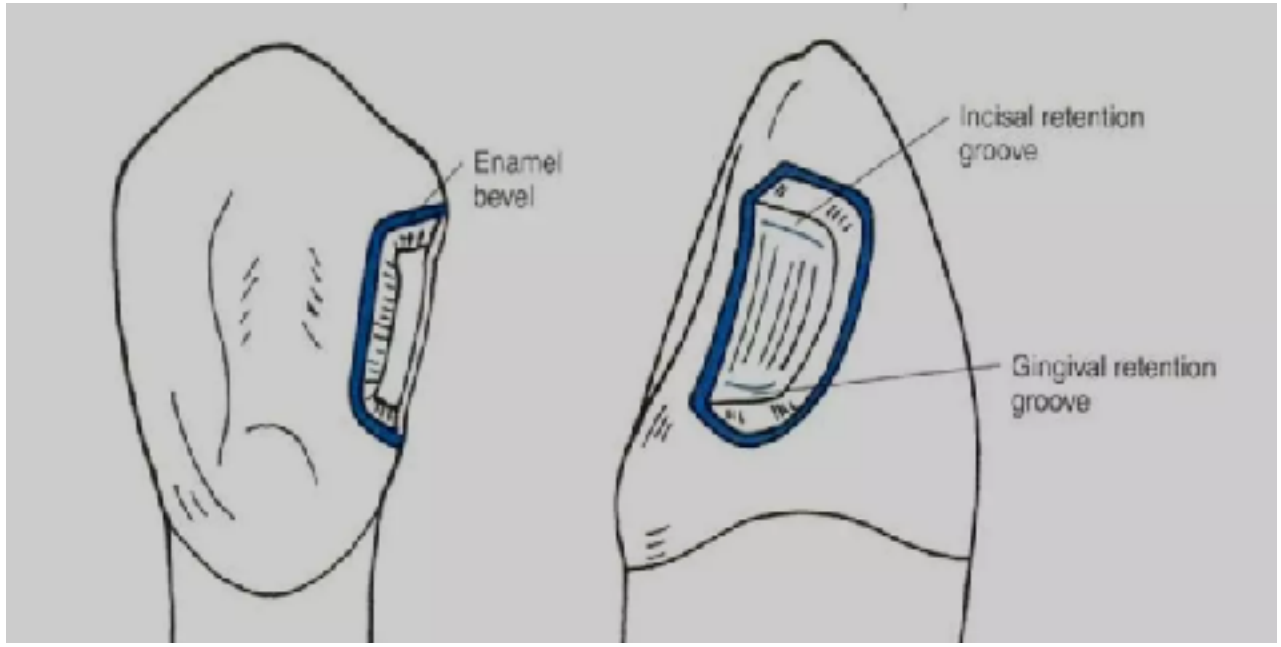
A. Residual caries
B. Recurrent caries
C. Forward caries
D. Backward caries

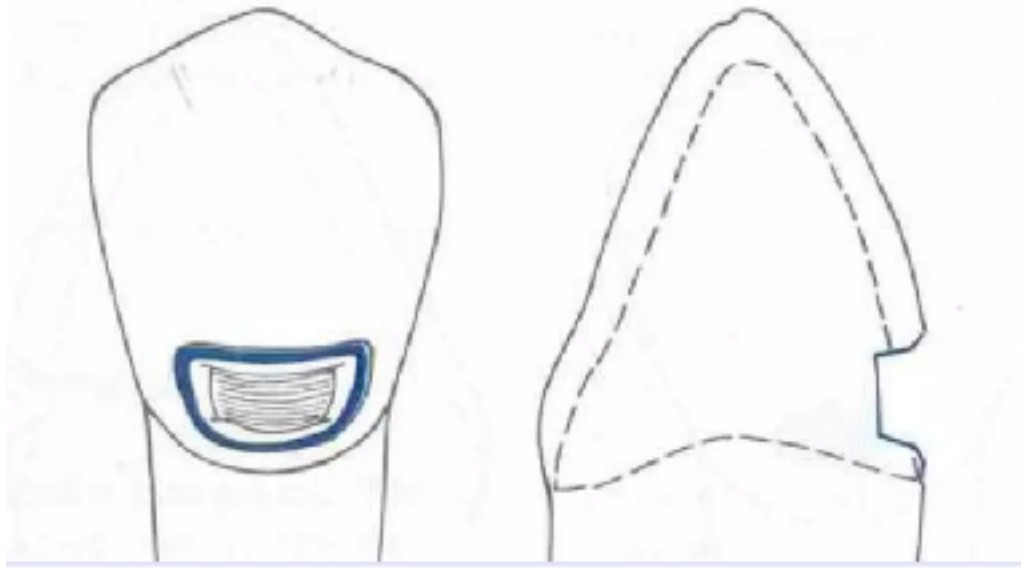


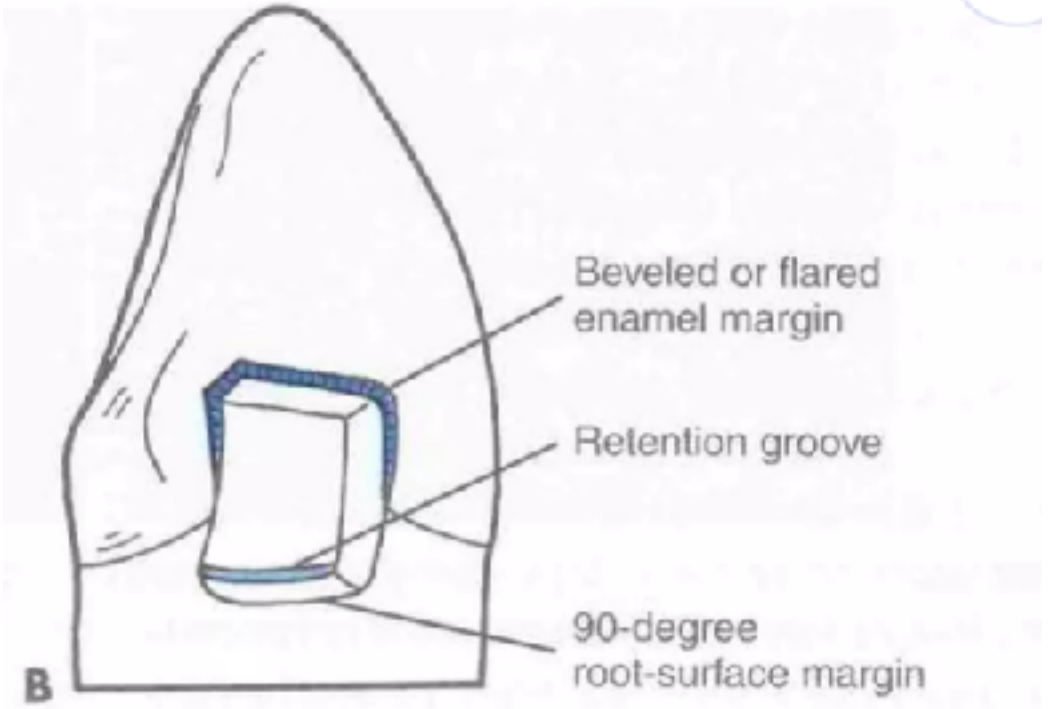
FEB-D

18. All of the following are characteristics of a Class V composite preparation except one. Which one is the exception?

A. All internal line angles should be rounded
B. Retention provided by undercuts made into the axial walls
C. Cavosurface margins when placed on enamel should be beveled
D. Outline form of the prep is determined by the extent of carious lesion
E. A buttjoint is not indicated when finishing the preparation on root surfaces


Enamel bevel
Incisal retention groove
Gingival retention groove




Beveled or flared enamel margin
Retention groove
90-degree root-surface margin
B

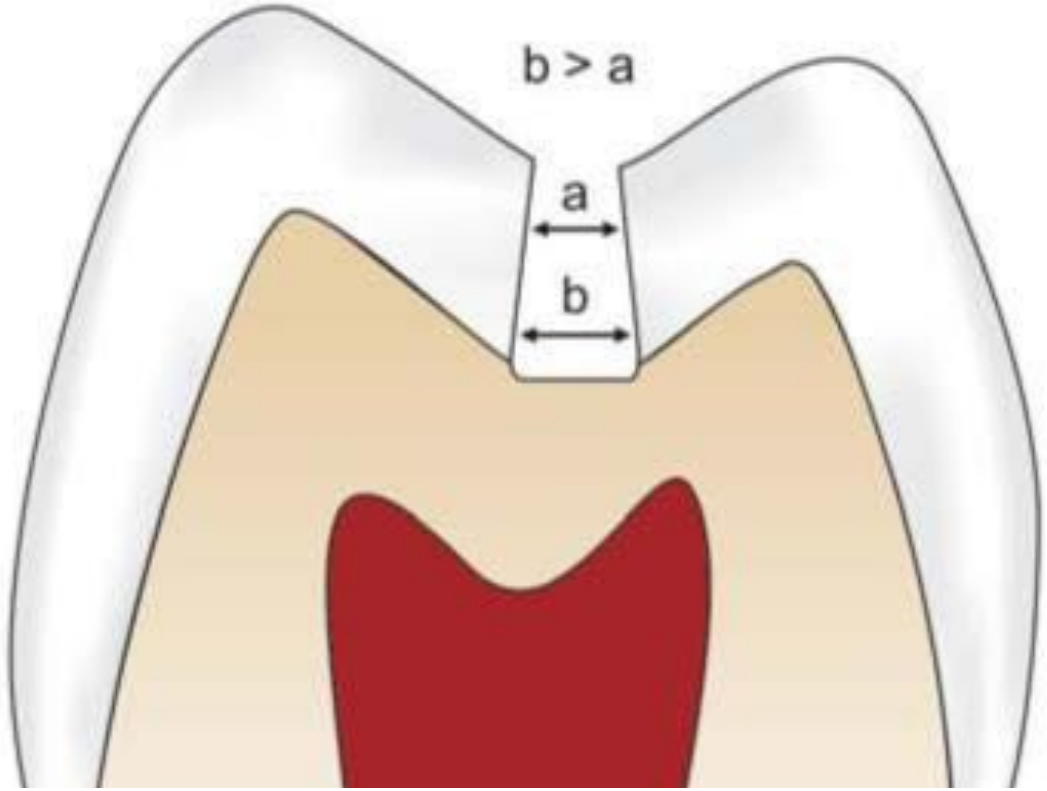
19. In a class I cavity prepared for silver amalgam, the direction of the buccal and lingual wall is:

A. Diverging

B. Converging

C. Parallel

D. Independent

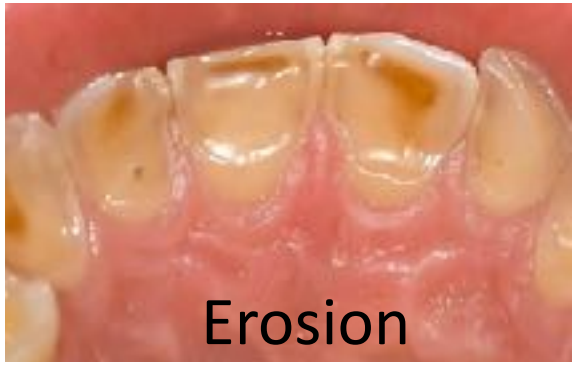


20. Initiation of caries by Streptococcus mutans is by the production of

A. Dextranase and soluble dextran
B. Insoluble dextran and glucosyl transferase
C. Soluble dextran and glycosyl transferase
D. None of the above

21. A 20-year old female presents for her initial dental examination and prophylaxis. During the examination, you observe wear on the occlusal surfaces of many posterior and anterior teeth. The wear on the posterior teeth are cup like and smooth, and on the anterior, it has smoothed out the lingual anatomy of the incisors. This is most typical of which type of wear?

A. Attrition
B. Erosion
C. Abrasion
D. Abfraction
E. Bruxism



Erosion

Attrition

22. An instrument (15-89-9-12) for bevelling axiopulpal line angle is called

_____

A. Enamel hatchet
B. Finishing bur
C. Hoe
D. Gingival margin trimmer

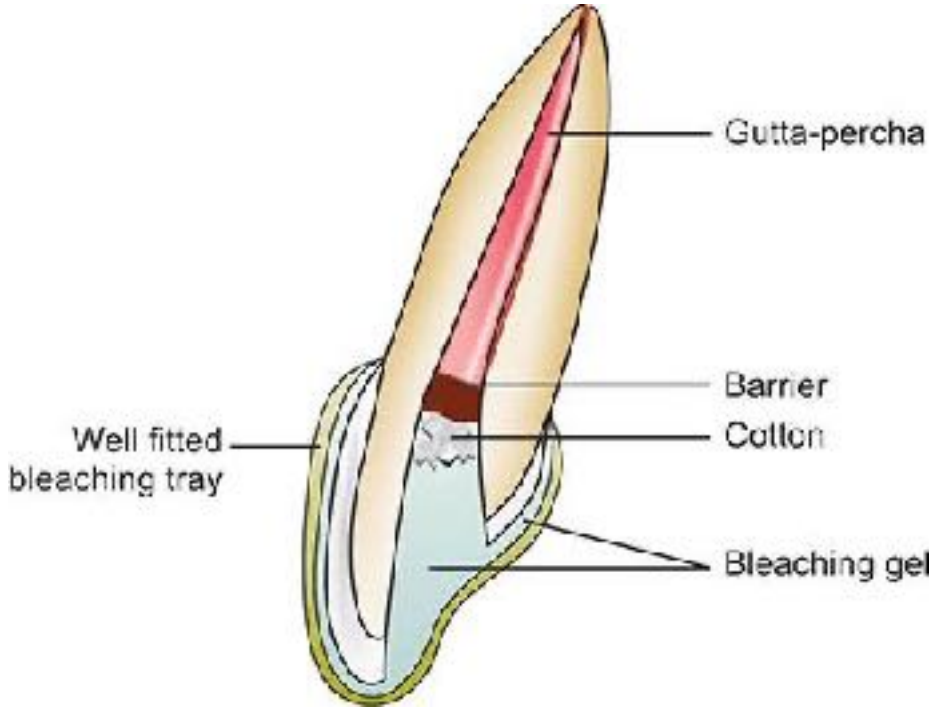
23. What is the ideal amount of remaining marginal ridge thickness for a GV black cavity preparation?

A.1.0mm
B.1.6mm
C.1.5mm
D.2.0mm
E.2.5mm

Resistance form
1.6mm for premolars, 2.0mm for molars

24. Which of the following is the most likely postoperative complication of bleaching a tooth that has not been adequately obturated?

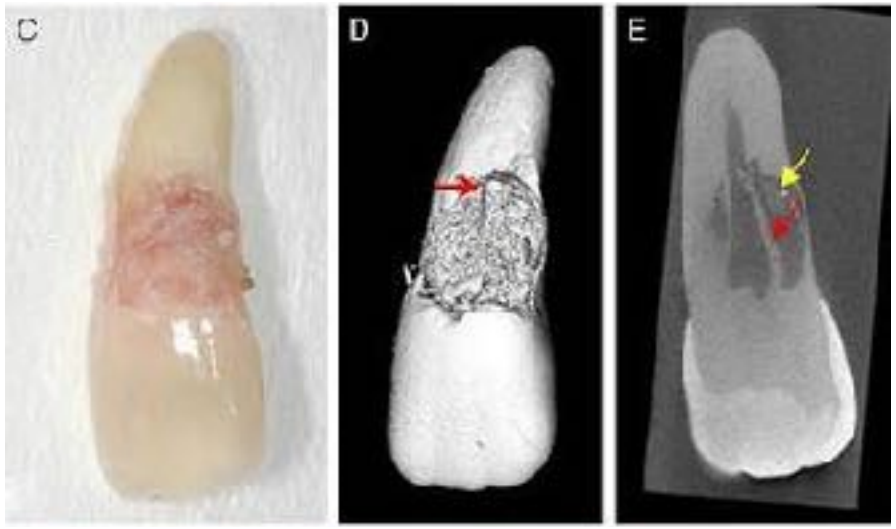
A. Fracture
B. Discoloration
C. Acute apical periodontitis
D. Chronic apical periodontitis
E. External cervical root resorption



2 most common complication of bleaching on non vital tooth

A. Apical periodontitis — Non-vital (improperly sealed/improperly obturated)

B. Cervical resorption — Non-vital (highly concentrated h202 diffusing through dentinal tubules to the cervical cementum)



25. Bevelling of the full thickness of enamel wall, but no bevel on dentin is a

A. Partial bevel "Ultrashort bevel", less than 2/3 of enamel

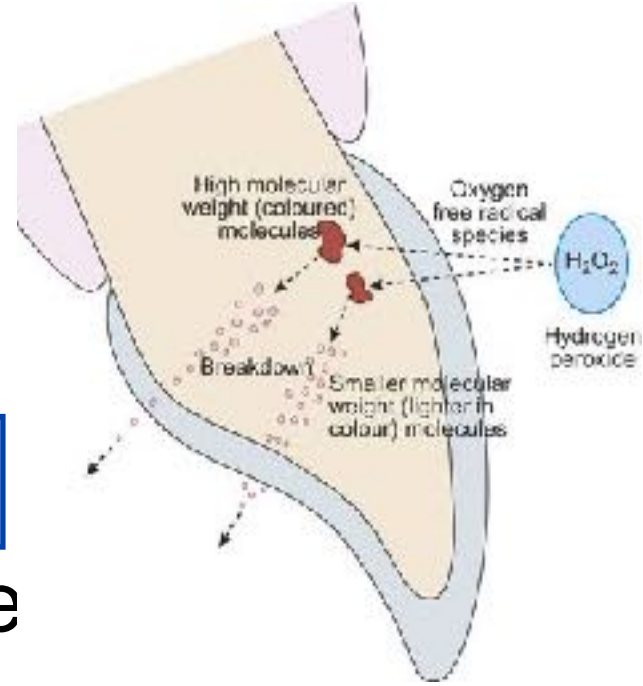
B. Full bevel Full enamel and full dentin

C. Short bevel Full enamel, but no dentin

D. Long bevel Full enamel, and 1/2 or less than 2/3 of dentin

26. In bleaching of dentin in endodontically treated teeth, composite resin fails to polymerize upon placement on a tooth surface that has undergone bleaching treatment. What is the possible reason for this non polymerization of the composite resin?

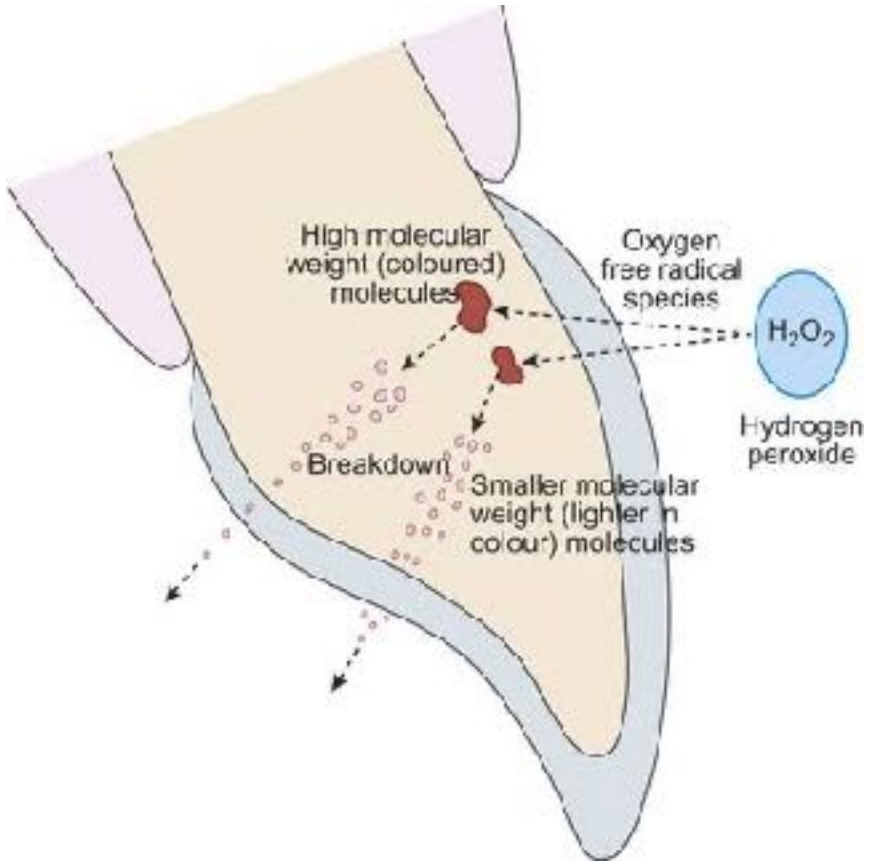
A. Decrease oxygen content in dentinal tissue
B. Presence of zinc oxide eugenol
C. Increase oxygen content in dentinal tissue
D. Absence of oxygen content in dentinal tissue



Teeth whitening

Reactive oxygen species cause the breakdown of larger pigmented molecules into smaller less visibly pigmented molecules

Oxidation-reduction



Post-bleaching restoration of access cavity

-residual oxides from bleaching affect bond strength of composite and interfere with the polymerization

-wait for 1-2 weeks after bleaching

-use sodium ascorbate to enhance removal of residual oxygen

27. What is the wavelength of visible light used for curing light cure resin restorations?

A.400-420 nm

B.420-475 nm

C.300-370 nm

D.475-500 nm

Light source:
*quartz-tungsten-halogen lamps
*Light emitting diode (LED) lamps
*plasma arc curing lamps
*argon laser lamps

Light is absorbed at **420-470 nm**

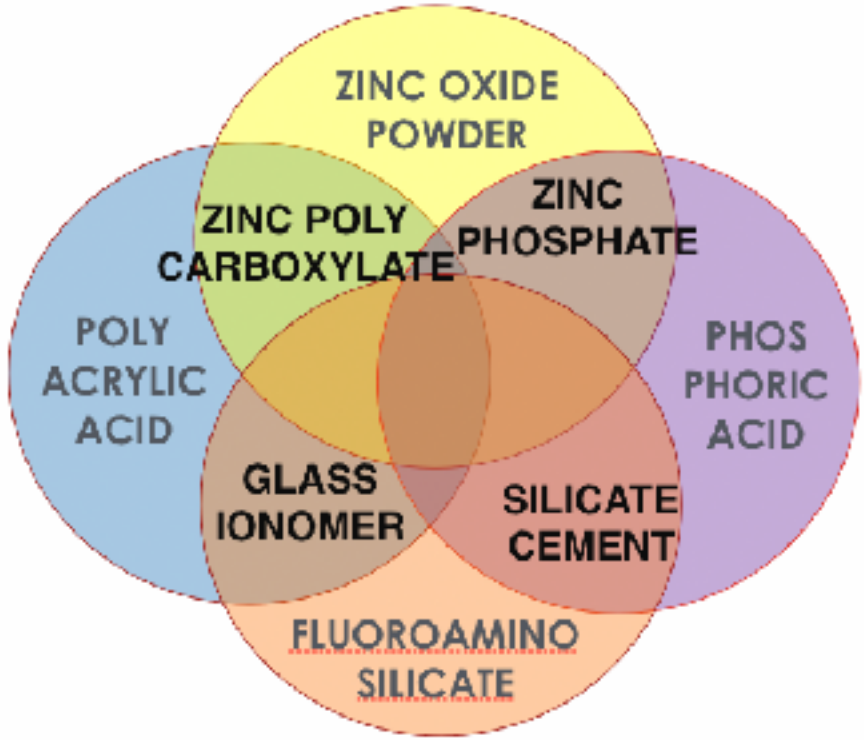
1-2mm ideal distance of light curing machine to composite (**less than 6mm)**

How many mm per increment to ensure sufficient curing?

2mm

28. Which of the following shows chemical bond with enamel?

A. Composites
B. Direct filling resins
C. Polycarboxylate cement
D. Bis-GMA resin in PFS



29. In air-abrasive technology, the abrasive particles used are

A. Iron oxide
B. Tin oxide
C. Aluminum oxide
D. Sodium bicarbonate

Air abrasion

Aluminum oxide powder

Air polish/Air flow

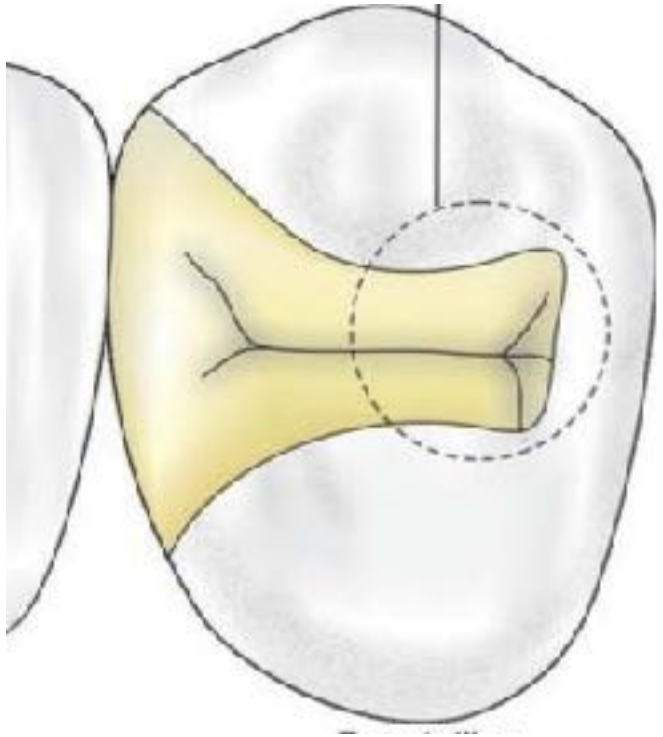
-Sodium bicarbonate powder
-Glycine powder
-zirconium silicate powder

30. For most effective cutting and long usefulness of a tungsten carbide bur, it should be

A. Rotating slowly before contacting the tooth
B. Rotating rapidly before contacting the tooth
C. Placed in contact with tooth before starting
D. Rotating rapidly before entering in to the oral cavity

31. Which of the following allows proper retention to prevent proximal displacement for a disto-occlusal class II gold inlay?

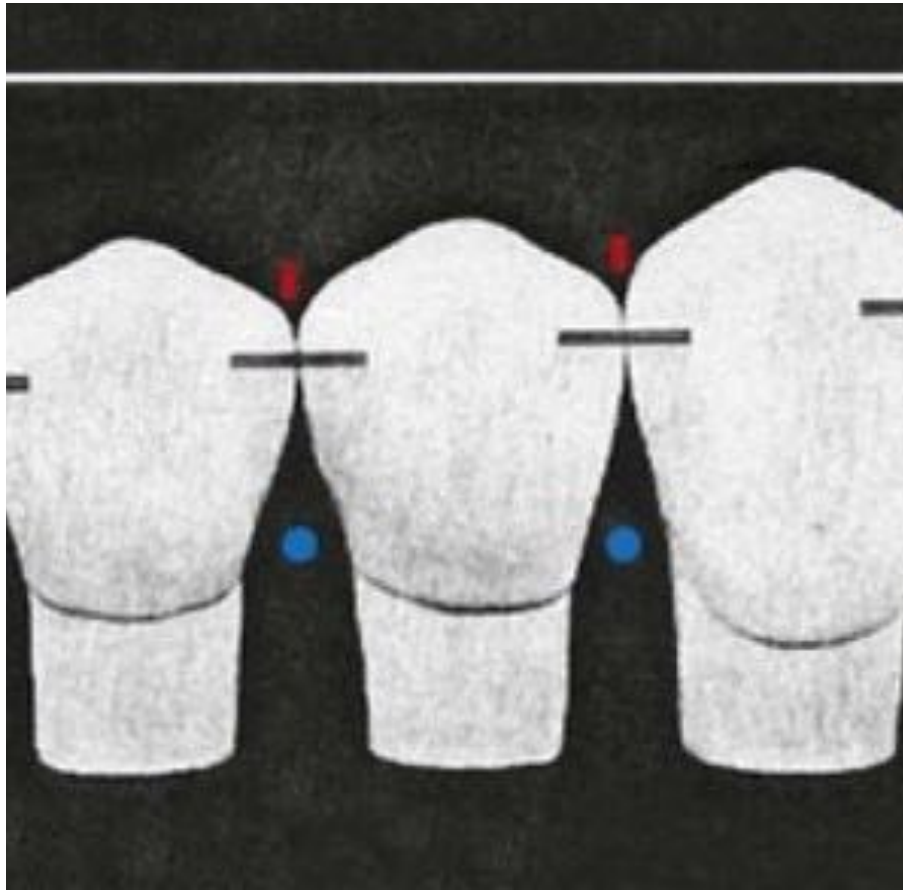
A. Undercut on mesial
B. Undercut on buccal and lingual walls
C. Occlusal lock
D. None of the above



Dovetail / "occlusal lock"

32. An incipient carious lesion on an interproximal surface is usually located

A. At the contact area
B. Facial to contact area
C. Gingival to contact area
D. Lingual to contact area



33. The configuration factor ["C" factor] for an occlusal class I cavity is

- ***ratio of bonded surfaces to the unbonded surfaces***

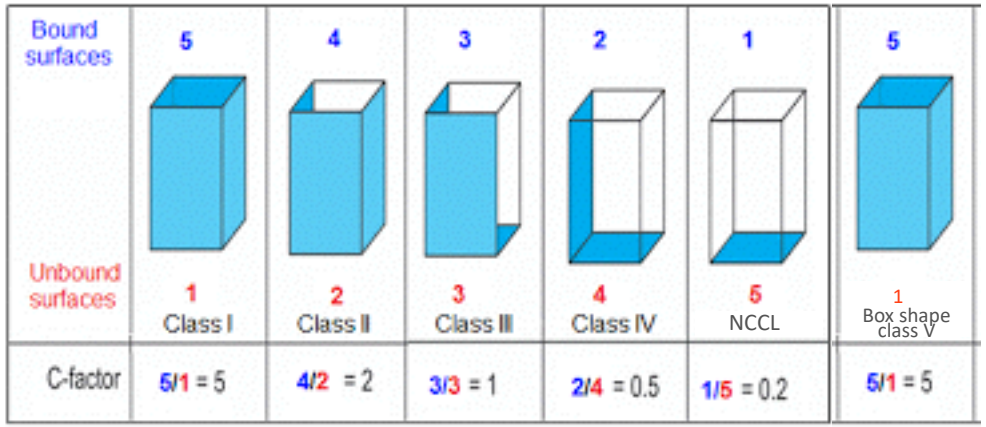
A.2

B.3

C.4

D.5

C-FACTOR
5:1 (5)
4:2 (2)
3:3 (1)
3:3 (1)
2:4 (0.5)
5:1 (5)
1:5 (0.2)



- ***Traditional "box" class V cavity***
- ***Non-carious cervical lesion (abrasion/abfraction)***

REMEMBER: Non-carious cervical lesions (NCCL)

Abfraction

-heavy eccentric occlusal force leading to tooth flexure & formation of microfractures on CEJ

APPEARANCE: *wedge shape or V-shaped notch (acute angle)w/ sharp internal & external line angles in cervical 3rd*

Abrasion

-direct friction forces between teeth & external object (ex: toothbrush)

APPEARANCE: *rounded notch in the cervical 3rd*

34. Which of the following prevents fracture of MO amalgam:

A. Bevelling axiopulpal line angle
B. Occlusal dovetails
C. Pin retainer amalgam
D.Removal of unsupported enamel rods

35. An ion-releasing composite material that reduces secondary caries formation at the margin of a restoration by inhibiting bacterial growth, in response to decrease in the pH

A. Ceromers
B. Ormocers
C. Smart composites
D. Compomers

36. In silver modified ART technique (SMART), silver is added in which form?

A. Silver fluoride

B. Silver diamine fluoride

C. Silver nitrate

D. Nano silver particle

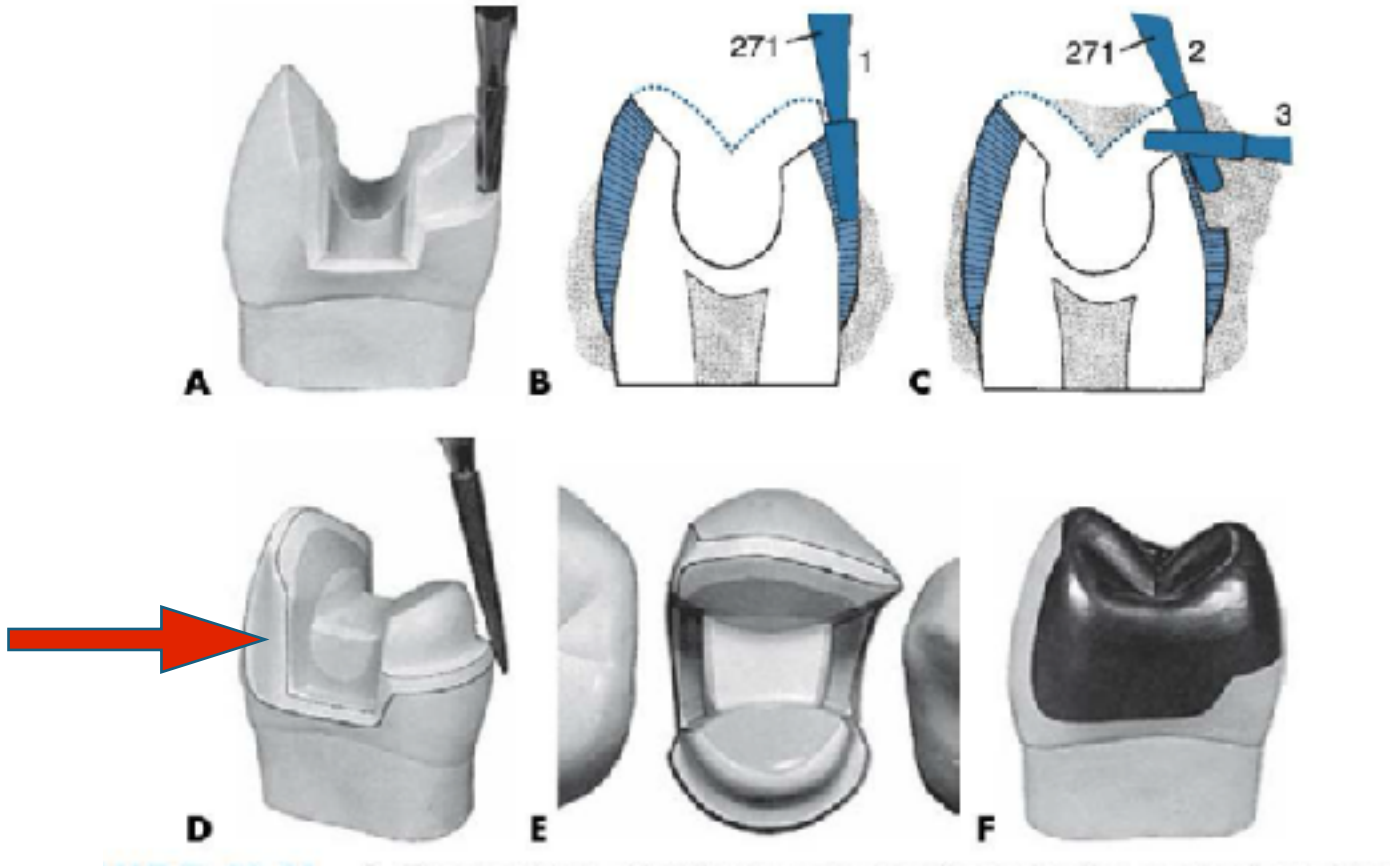
SMART

"SILVER MODIFIED ATRAUMATIC RESTORATIVE TREATMENT"

DMF DENTAL TRAINING CENTER BY: DR. C. GUERRERO-PINEDA

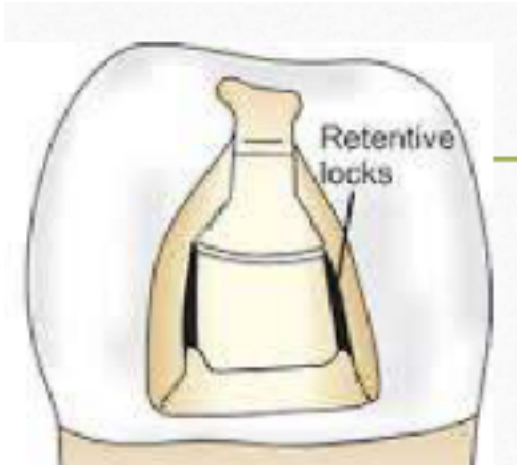
37. Thin extensions of facial and lingual proximal margins in an onlay preparation for teeth that exhibit split tooth syndrome

A.Collar
B.Slot
C.Bevel
D.Skirt



SKIRT & COLLAR

LOCK. SLOT. COVE

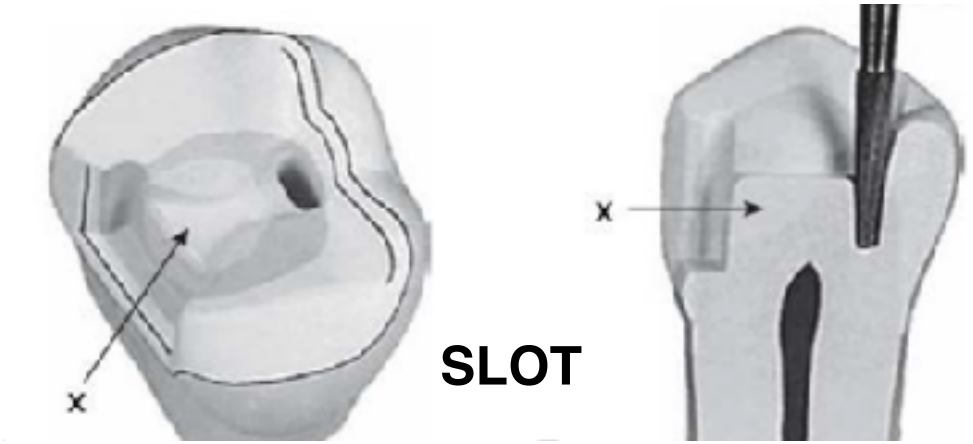


LOCK

(Longitudinal/vertical retention groove)
(Location: walls)

COVE

(Conical depression)
(Location: floor & walls)



SLOT

(Transverse/horizontal retention groove)
(Location: floor)

38. It is essential to lubricate dam before applying it. Which of these is NOT a suitable lubricant?

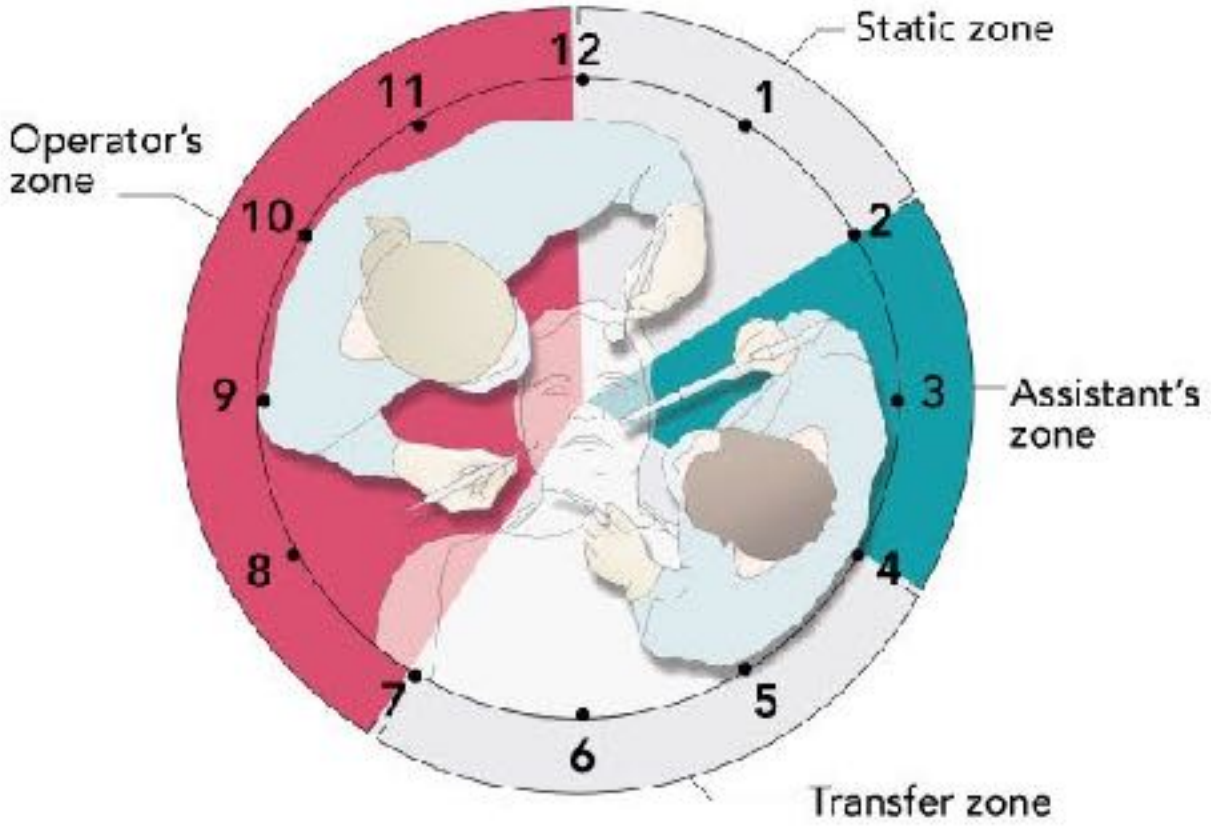
A. Shaving cream
B. Liquid soap
C. Scrub gel
D. Vaseline

STEPS FOR PLACING RUBBER DAM:

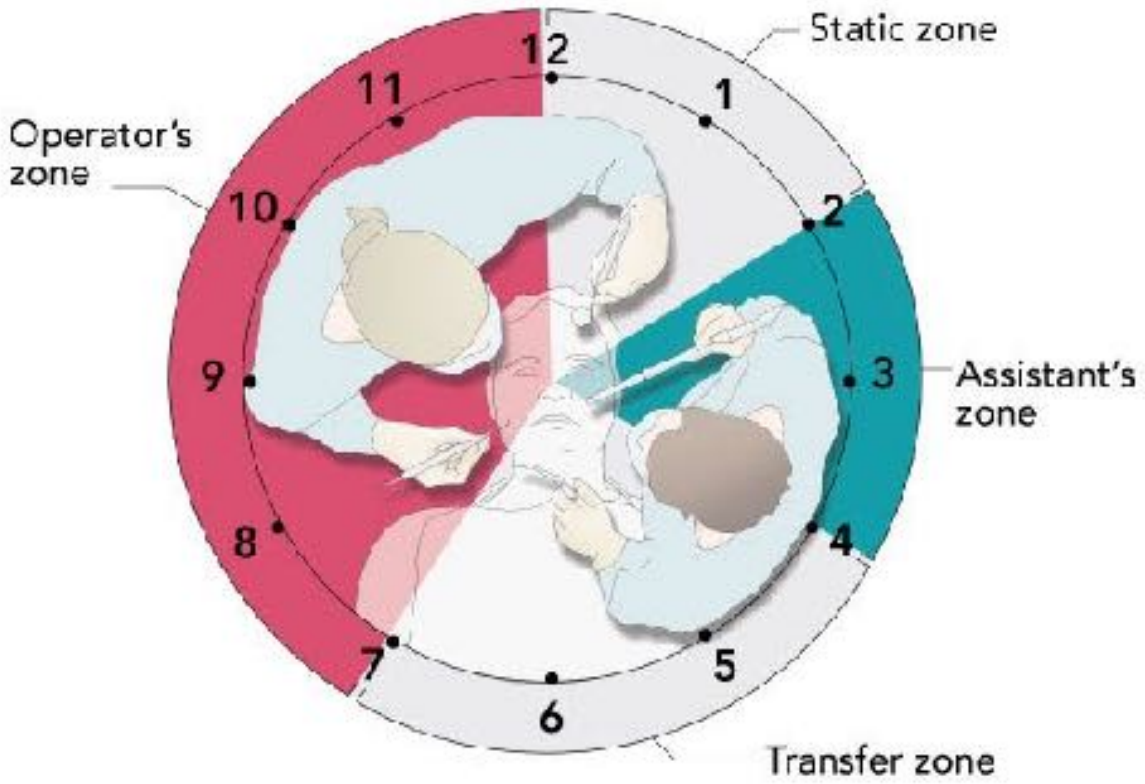
1. *Testing and lubricating the proximal contacts*
2. *Punching the holes*
3. *Testing the retainer stability and retention*
4. *Positioning the dam over retainer*
5. *Attaching the frame*
6. *Passing the dam through posterior contact*
7. *Passing the septa through the contacts*

39. From the following, which is not a zone in four-handed dentistry?

A. Operator zone
B. Transfer zone
C. Patient zone
D. Assistant zone
E. Static zone

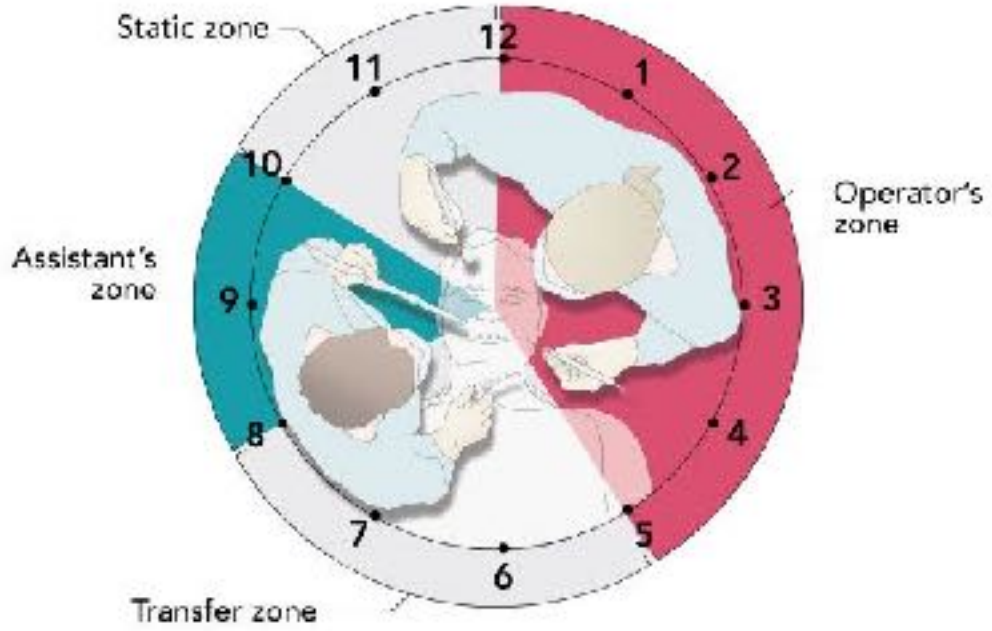


Four-handed dentistry
(sit-down position)



Right-Handed Dentist		
Operator's zone	7 to 12 o'clock	8 to 11
Transfer zone	4 to 7 o'clock	5 to 8
Assistant's zone	2 to 4 o'clock	2 to 5
Static zone	12 to 2 o'clock	11 to 2

11 o'clock : considered the most universal operating position for dentist



Left-Handed Dentist		
Operator's zone	12 to 5 o'clock	1 to 4
Transfer zone	5 to 8 o'clock	4 to 7
Assistant's zone	8 to 10 o'clock	7 to 10
Static zone	10 to 12 o'clock	10 to 1

40. Which of the following is the main constituent of porcelain

A. Feldspar 60-80%

B. Kaolin 3-5%

C. Quartz 15-25%

D. Clay

41. Cervical restorations fail mostly due to

A. Inadequate moisture control
B. Marginal gingivitis
C. Cuspal flexure
D. All of the above

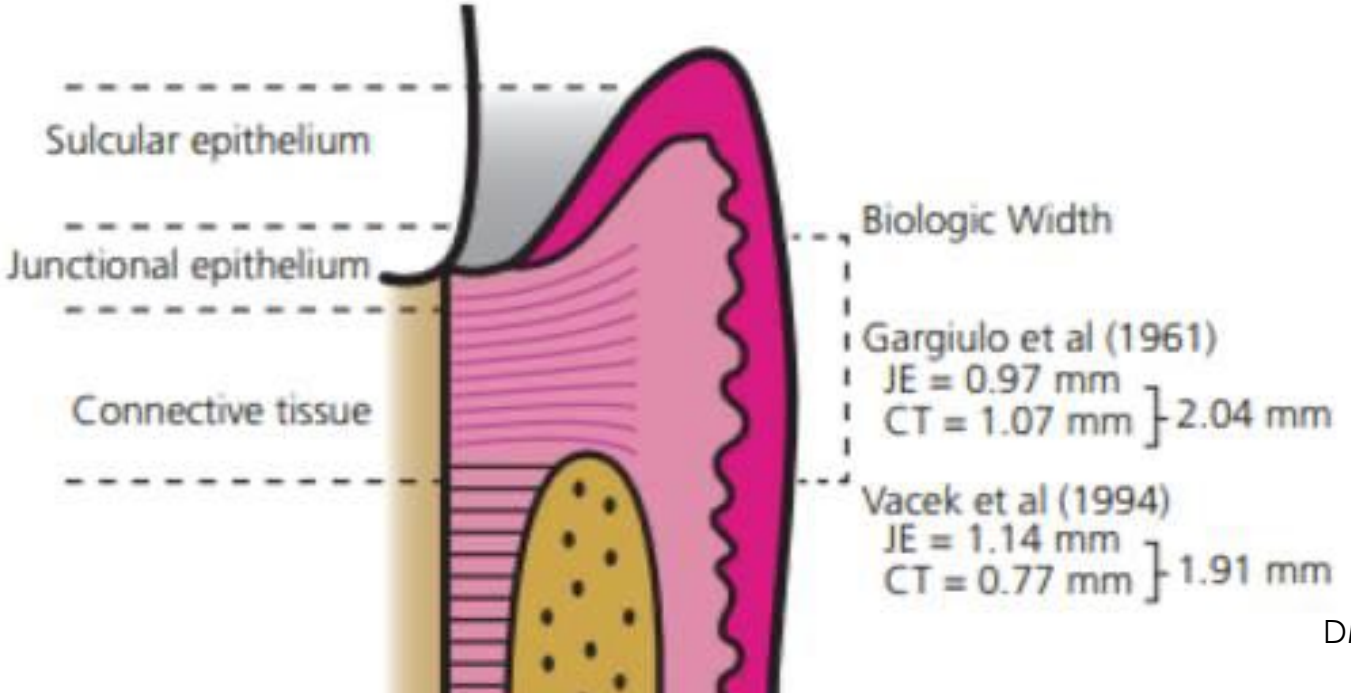
42. Gingival extension for a restoration should be:

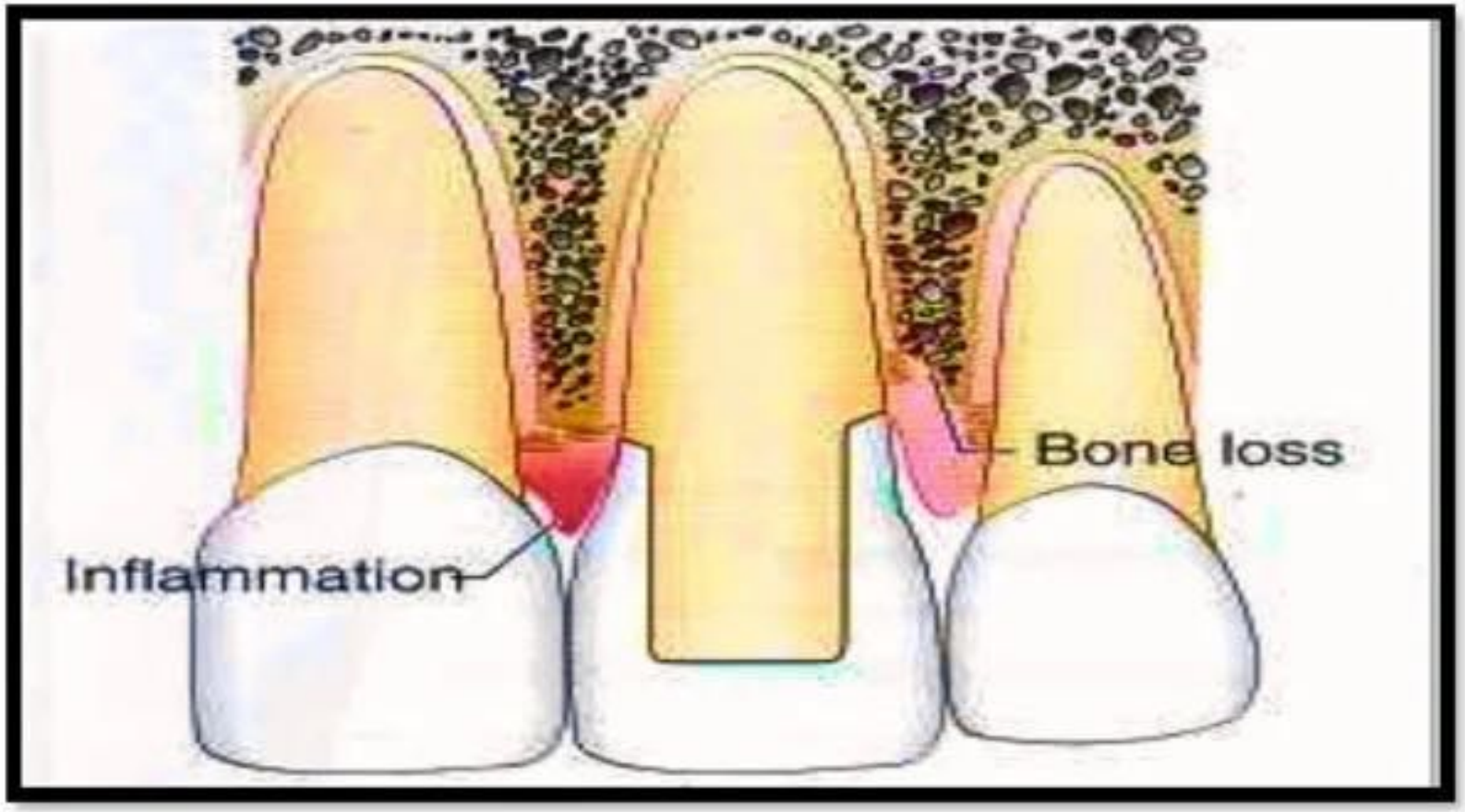
A. At the gingival crest

B. At least 1mm above the alveolar crest

C. At least 3mm above the alveolar crest

D. Doesn't have any relation with alveolar crest height




Bone loss
Inflammation

43. Retraction cord is often used with a hemostatic agent to capture an accurate impression. Which hemostatic agents minimize tissue damage?

A. $Fe_2(SO_4)_3$ and $AlCl_3$

B. $ZnCl_2$ and $AlCl_3$

C. $Fe_2(SO_4)_3$ and $ZnCl_2$

D. $(SO_4)_2$ and $ZnCl_2$

E. Epinephrine and $Fe_2(SO_4)_3$

Zinc chloride can cause tissue necrosis!!

44. What is the least invasive treatment for mottled enamel?

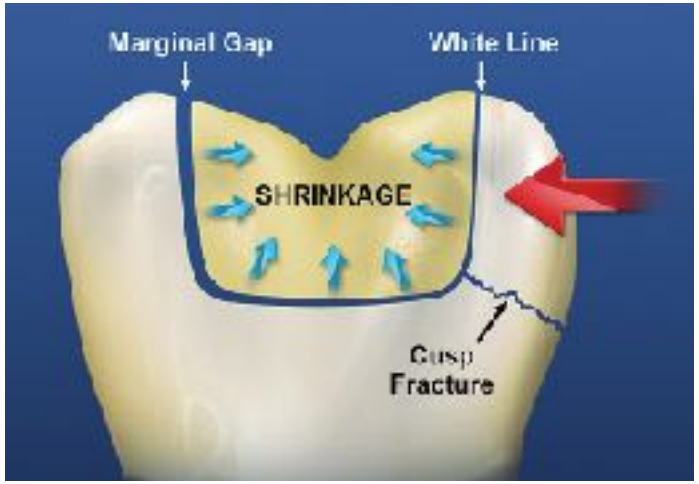
A.Zirconia veneers
B.Porcelain veneers
C.Composite bonding
D.Microabrasion

6% hydrochloric acid

Fig. 11: Active etching with 15 % hydrochloric acid (Icon Etch, DMG) using the special vestibular tips.

45. The typical polymerization shrinkage of resin composite is

A.2-3%
B.4-5%
C.6-7%
D.8-9%

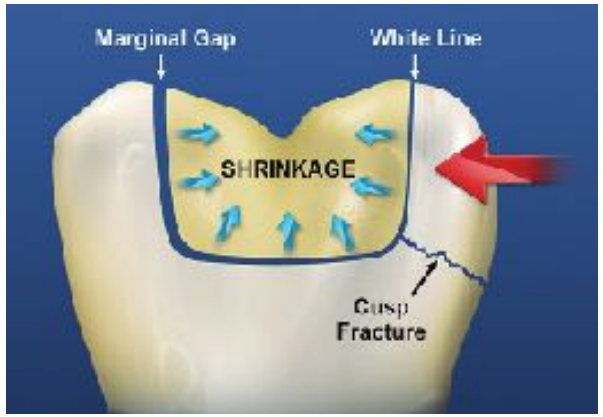


2.2 to 2.4%

or 2-3%

POLYMERIZATION SHRINKAGE:

- the greatest DISADVANTAGE OF POLYMERIZATION



2.2 to 2.4% (Sturdevant)

or 2-3% (BEQ question banks)

- **Faster polymerization** = MORE SHRINKAGE
- **Lesser filler** = MORE SHRINKAGE
- **Darker Shade** composites **NEEDS LONGER CURING TIME**

46. What is the current recommended bleaching material for the “walking bleach” technique?

A.5.25% sodium hypochlorite -for irrigation during RCT

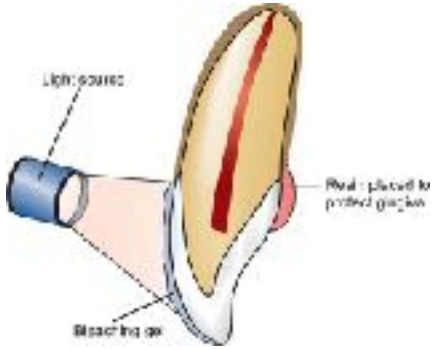
B.30% hydrogen peroxide -in-office for vital & nonvital bleaching

C.Sodium perborate -at-home/walking bleaching for nonvital tooth (carbamide peroxide for vital&nonvital at home/walking)

D.Iodoform -antiseptic dressing for dental surgeries, extractions, and RCT

E. A & b only

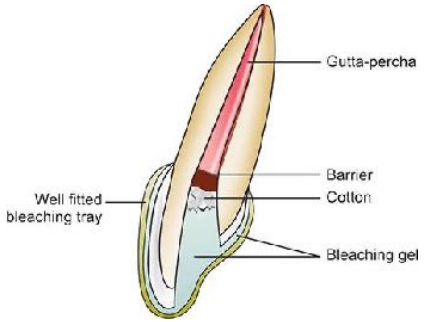
VITAL BLEACHING
"POWER BLEACHING"



1. IN-OFFICE BLEACHING:
35% HYDROGEN PEROXIDE

2. AT-HOME BLEACHING:
10-15% CARBAMIDE PEROXIDE

NON-VITAL BLEACHING

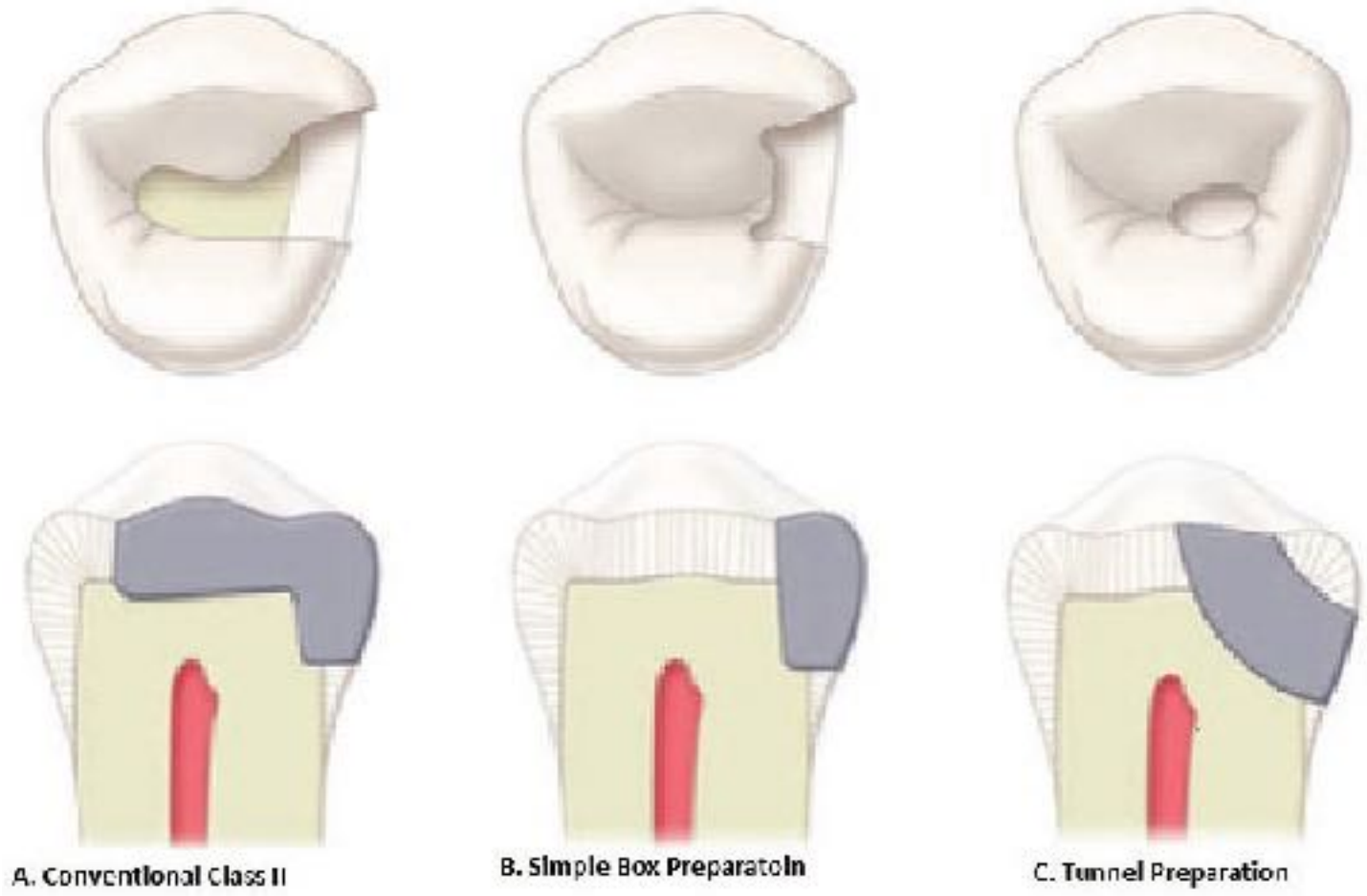


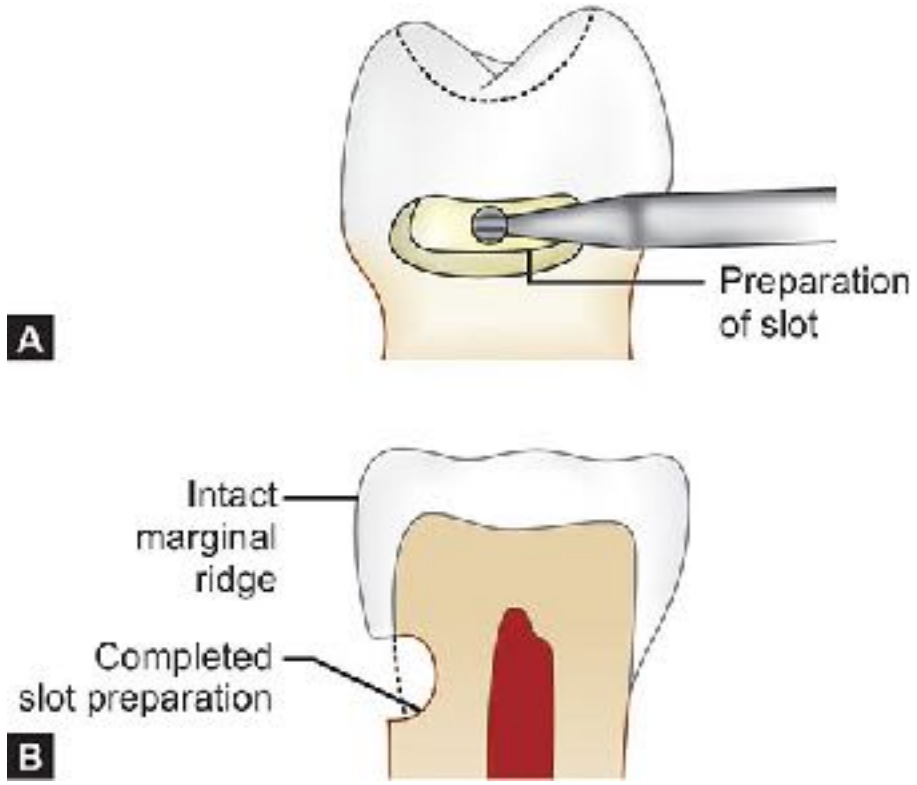
1. IN-OFFICE/NON-VITAL:
"THERMOCATALYTIC TECHNIQUE"
35% HYDROGEN PEROXIDE

2. AT-HOME/WALKING BLEACHING
10% CARBAMIDE PEROXIDE
OR SODIUM PERBORATE

47. Which of the following are/is acceptable types of tooth preparations to receive composite resins as restorative materials?

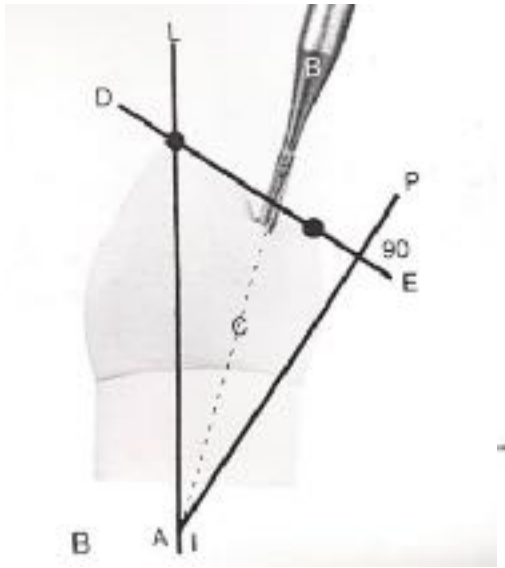
A. Conventional and bevelled conventional
B. Modified preparation where only the defect is removed
C. Box only or slot preparations
D. All of the above

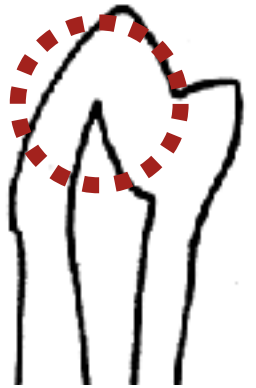

A. Conventional Class II
B. Simple Box Preparatoln
C. Tunnel Preparation


A
Preparation of slot
Intact marginal ridge
Completed slot preparation
B

48. While preparing occlusal cavity for amalgam in mandibular first premolar, the bur

A. Is kept absolutely vertical
B. Is tilted slightly buccally
C. Is tilted slightly lingually
D. Can be kept in any direction





49. For cast restorations the cavity wall should have a taper of

A.30-45

B.5-10

C.2-5

D.none of the above

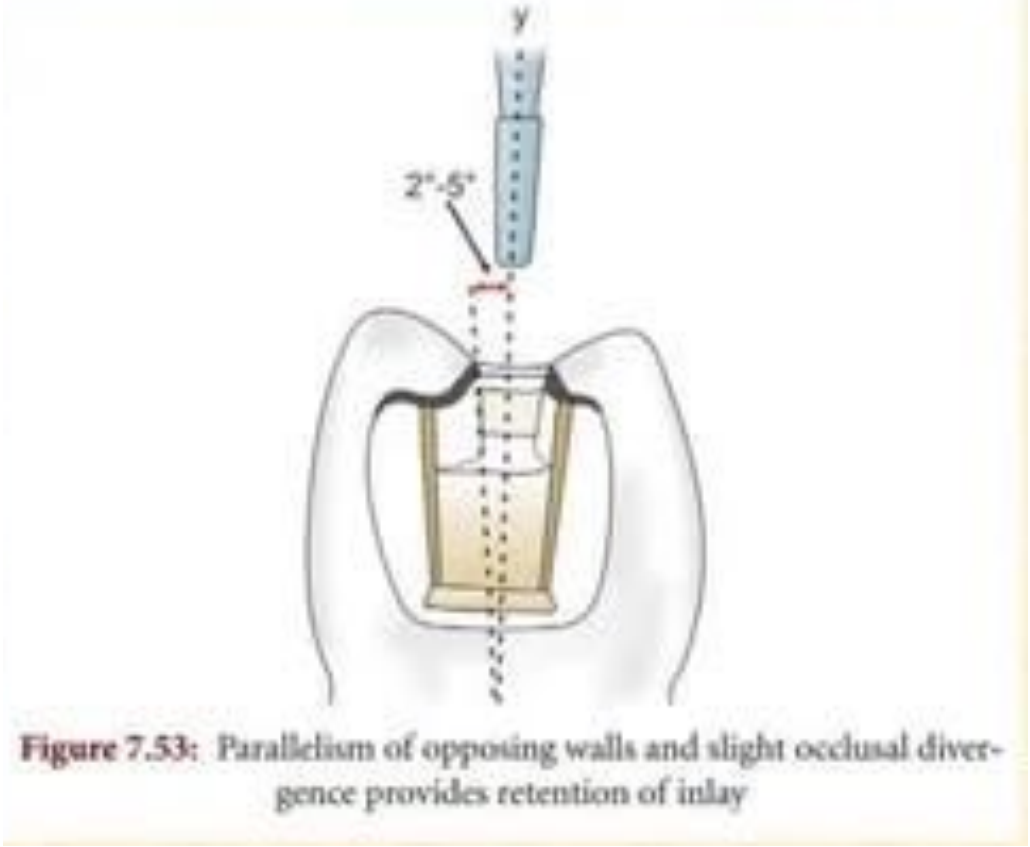


Figure 7.53: Parallelism of opposing walls and slight occlusal divergence provides retention of inlay

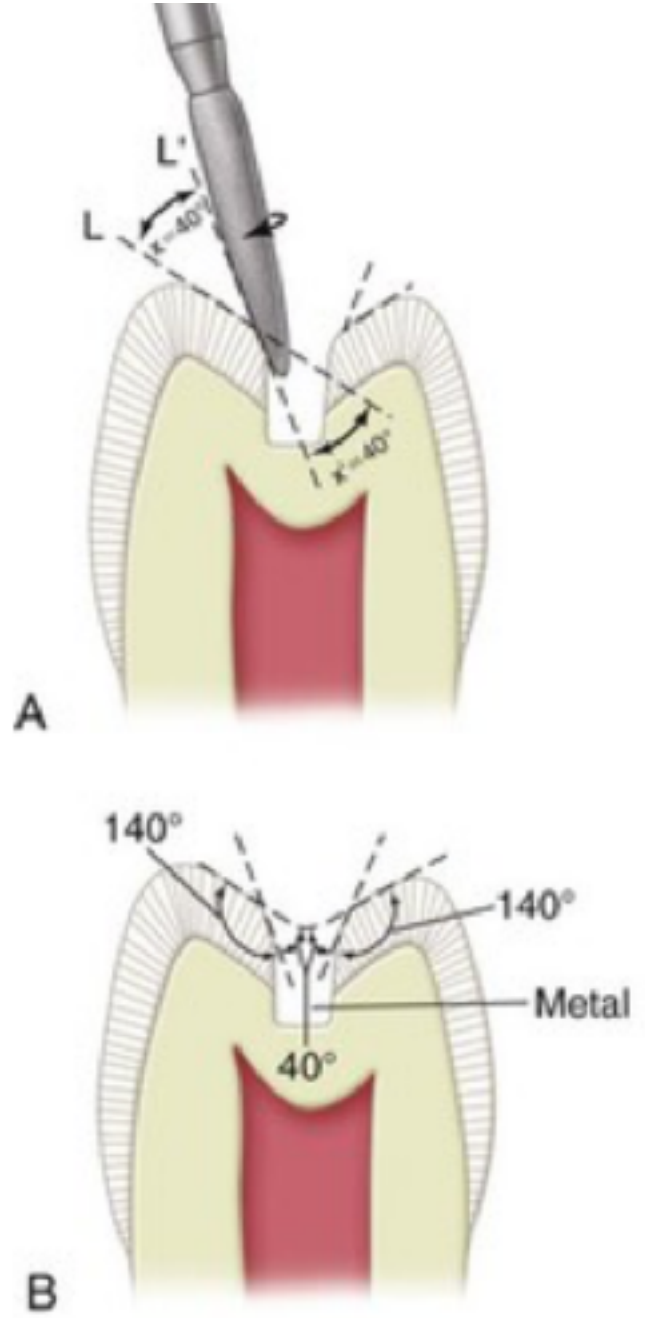
50. The cavosurface angle for inlay cavity preparation

A.90 degree

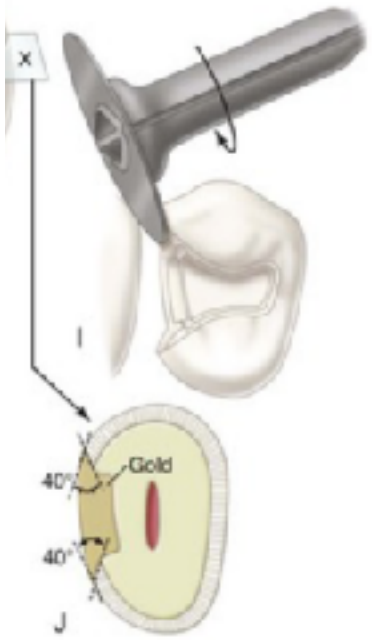
B.150 degree

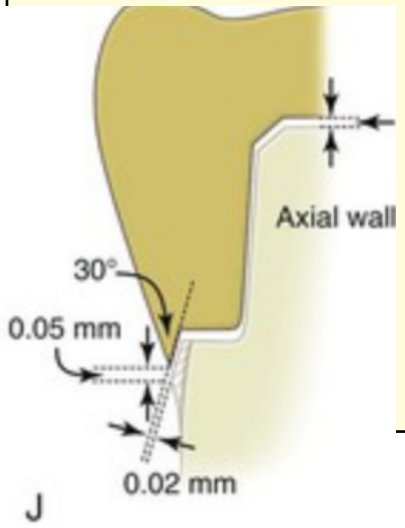
C.Less than 90 degree

D.180 degree



COMPARISON OF THE CAVOSURFACE MARGINS FOR A CONSERVATIVE CLASS II AMALGAM AND GOLD INLAY PREPARATION		
Characteristics	**Class II gold inlay**	**Class II amalgam**
Occlusal CSM	Beveled to result in 40-degree marginal metal ***(140-150 cavosurface angle)***	90 degree angle with the external surface
Proximal CSM	Extended up to the embrasures to prevent recurrent caries. Beveled to result in 40 degree marginal metal	90 degree angle with the external surface
Gingival CSM	Beveled to result in 30 degree marginal metal All 40 deg except Gingival CSM and Counterbevel of onlay	Beveled in permanent teeth (to produce 90-degree amalgam marginal angle (15-20 degrees) NO BEVEL IN DECIDUOUS!





51. All of the following cavity classifications can involve any teeth except for:

A. Class I
B. Class II
C. Class V
D. Class VI

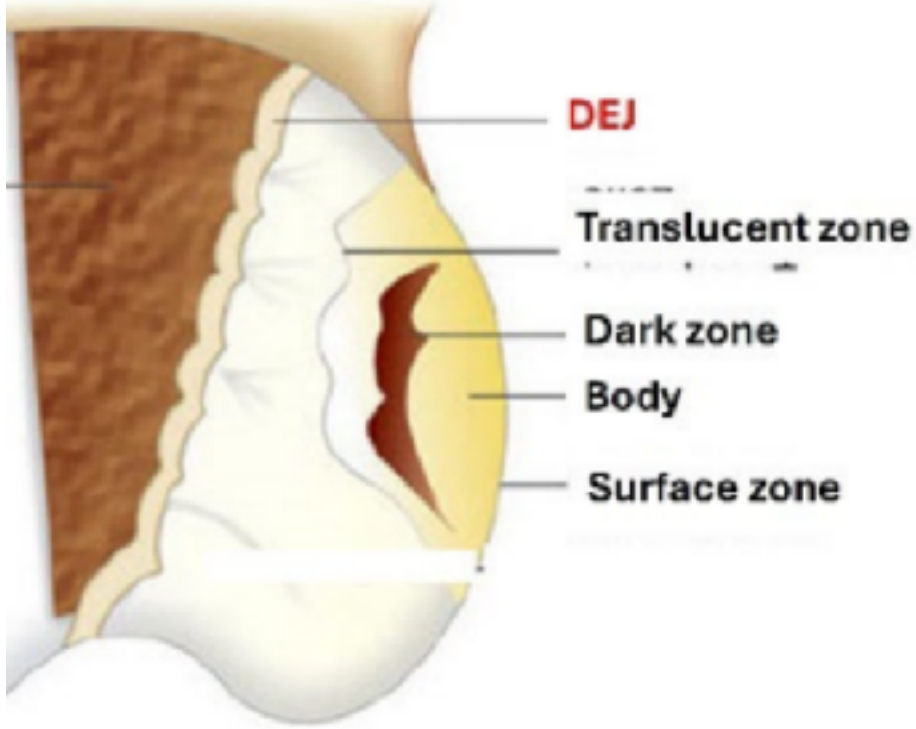
52. There are four zones of carious enamel, the zones of incipient lesions. Which of the four zones is the largest?

A. Translucent zone — Deepest zone

B. Dark zone — Zone of remineralization

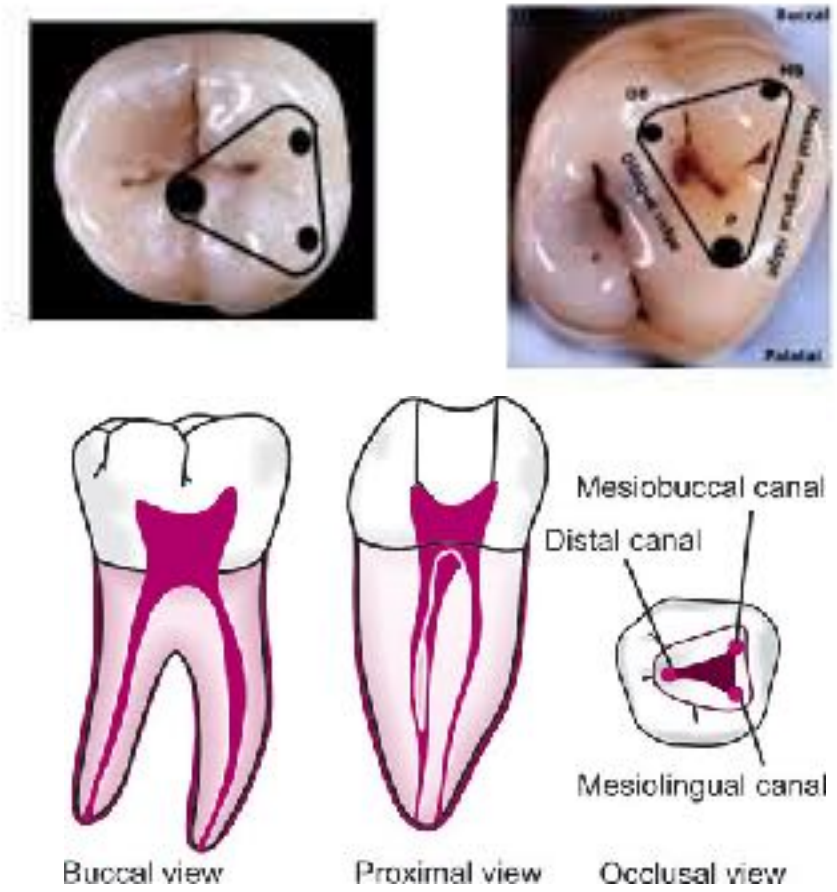
C. Body zone — Demineralized zone, largest

D. Surface zone — Outermost zone, unaffected by carious lesion



53. While preparing a Class II cavity on a maxillary first molar which of the pulp horns are likely to get exposed:

A. Mesiolingual and mesiobuccal
B. Distolingual and distobuccal
C. Mesiolingual and distobuccal
D. Distolingual and mesiobuccal



54. Which of the following cements should not be used to cement acrylic temporary crown:

A. Zinc oxide eugenol — Best choice for cementing acrylic temporary crown IF receiving non-resin permanent resto, contraindicated for temporization receiving composite inlay/onlay as permanent restoration

B. Zinc phosphate — Used for permanent luting crowns,inlays,fixed bridge (except ceramics), also insulating base for deep cavities

C. Glass ionomer — Contraindicated: CHEMICAL ADHESION

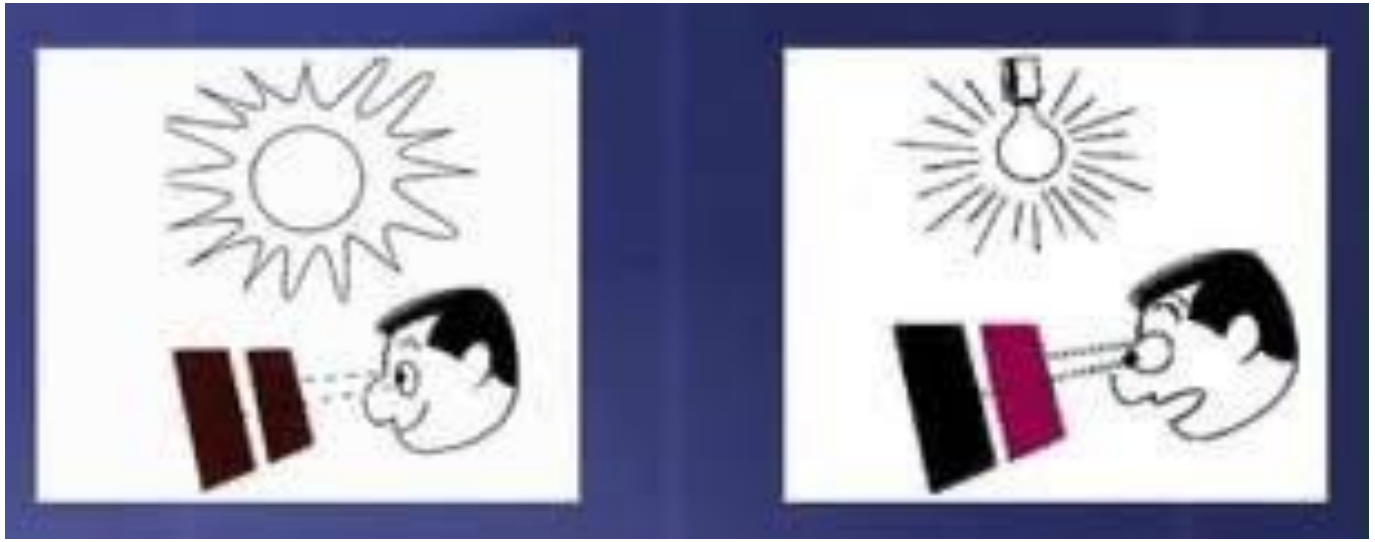
D. None of the above

55. The most common cariogenic bacteria responsible for coronal caries are:

A. Lactobacillus acidophilus
B. Actinomyces viscosus
C. Streptococcus sanguis
D. Streptococcus mutans and Lactobacilli

56. Phenomenon in which porcelain appears different under varying light condition is:

A. Translucency
B. Refractive optics
C. Metamerism
D. Opacification



57. In a patient with reduced salivary flow, the carious incidence is

A. More than the patient with normal salivary flow
B. Less than the patient with normal salivary flow
C. Unaffected
D. None of the above

Saliva washes the food away
(Bacterial clearance)

Adults produce
1-1.5 liters of saliva
per day

Reduced/Decrease viscosity will decrease caries susceptibility

Reduced salivary flow will increase caries susceptibility

58. All of the following are true for Class II amalgam preparations, except one. Which one is the exception?

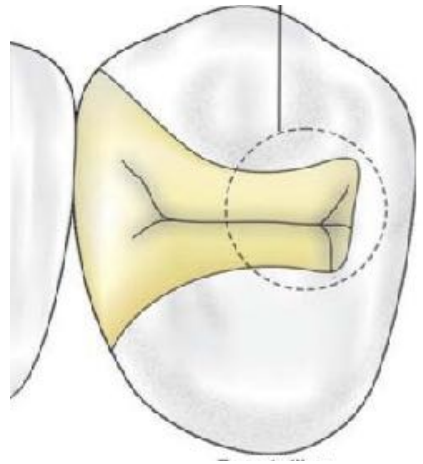
A. Proximal retention locks if placed, should be entirely in dentin, and are deeper gingivally than occlusally

B. The axiopulpal line angle is beveled or rounded to reduce stress

C. The uninvolved proximal wall should converge slightly toward the occlusal

D. The pulpal depth aids in the resistance form of the preparation

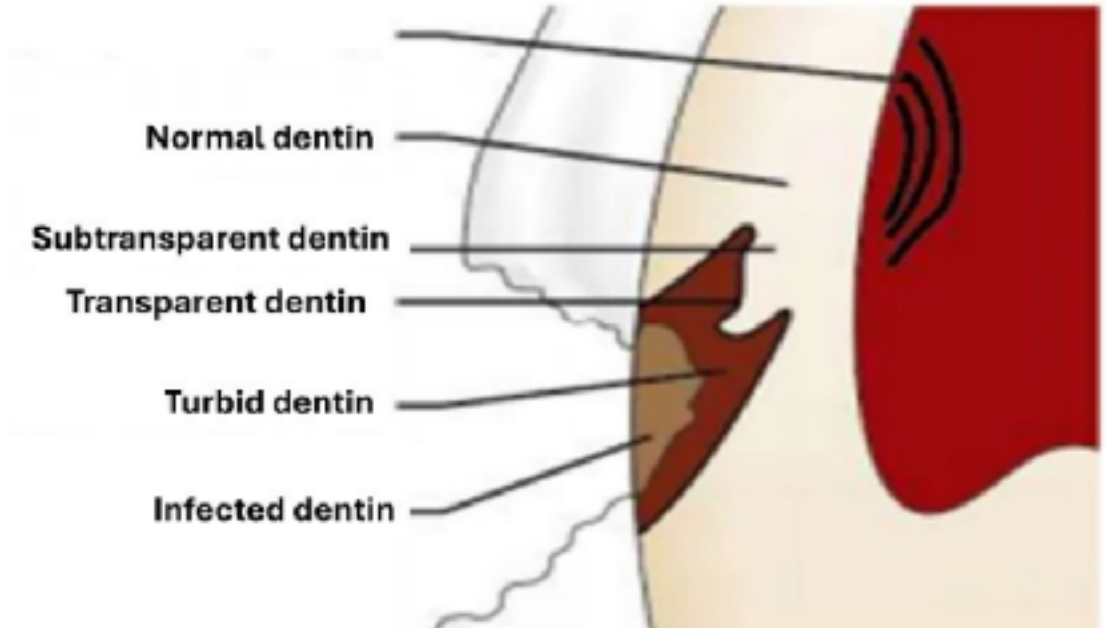
E. The buccal and lingual walls should converge to provide mechanical retention to the restorative material



Dovetail / "occlusal lock"

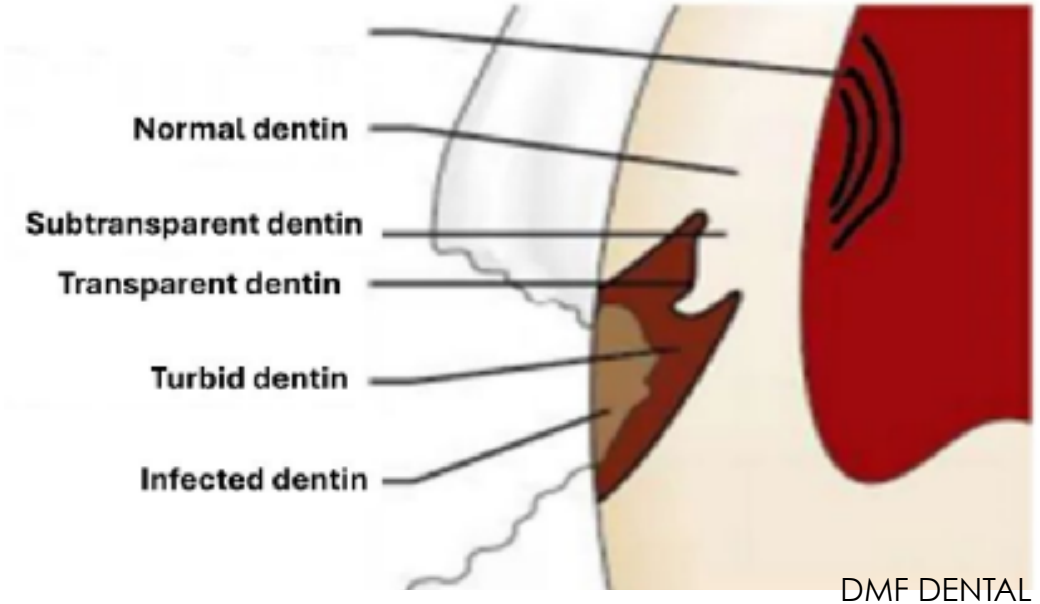
59. In a carious lesion in which caries has extended into dentin, which of the following zones contains infected dentin and is not capable of remineralization?

A. Normal dentin
B. Subtransparent dentin
C. Transparent dentin
D. Turbid dentin
E. Eburnated dentin



Zone of Carious Dentin

1) Normal Dentin – no bacteria present in this deepest unaffected area	
2) Subtransparent Dentin-zone of dentinal sclerosis, demineralization from acidogenesis, no bacteria found in tubules	Affected Dentin (dry leathery texture)
3) Transparent Dentin – demineralized softened dentin; but no bacteria in tubules	
4) Turbid Dentin – zone of bacterial invasion; presence of bacteria in tubules; must be removed during preparation	Infected Dentin (soft mushy texture)
5) Infected Dentin – zone of decomposed dentin, outermost carious zone with numerous microorganisms; must be removed	



60. Ceramic shade matching is an important step in achieving natural esthetics in dental tooth restoration. What is the most important dimension of color for dental restorations?

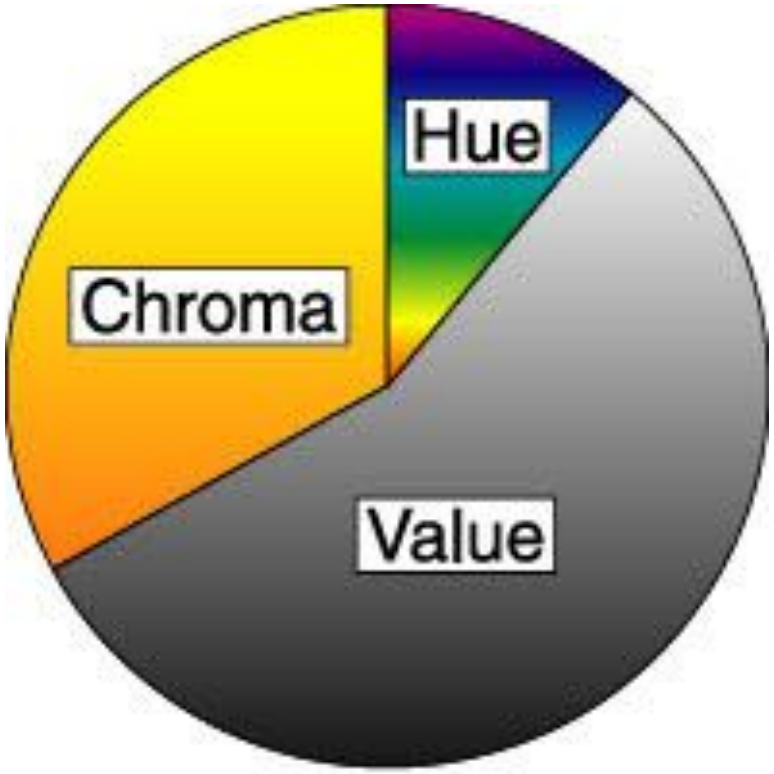
A. Color

B. Hue -words for describing the actual color: red, purple, blue

C. Value -lightness/darkness: overall intensity to how light or dark a color is

D. Chroma -saturation: strength or dominance of the hue

E. Metamerism



61. When removing a rubber dam. The first step is to:

A. Remove the clamp

B. Release the holder

C. Apply a water soluble lubricant

D. Cut the interseptal rubber with scissors

FIGURE 18. Stretching the rubber dam to expose the interseptal portion.

FIGURE 19. Cutting the interseptal portion of the dam.

FIGURE 20. Rubber dam ready for removal

Removal:

1. *Cut interseptal rubber*
2. *Remove retainer/clamp*
3. *Remove the dam*

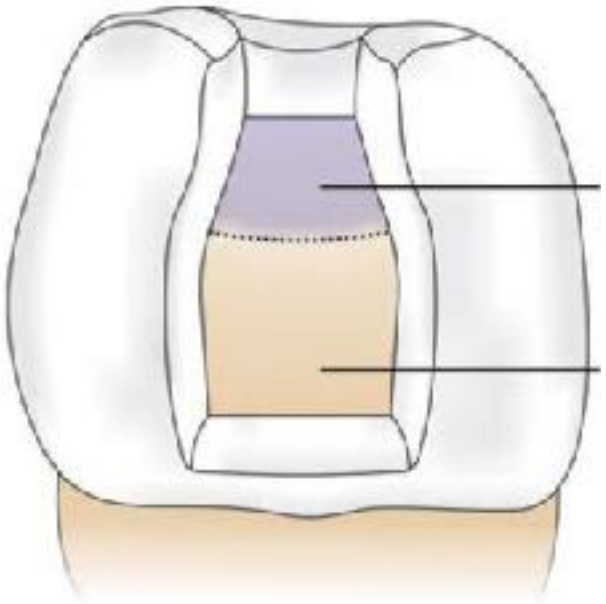
62. The total number of point angles present for Class V amalgam restorations is

A.4

B.6

C.8

D.11



CAVITY	LINE ANGLE	POINT ANGLE	C-FACTOR
Class I	8	4	5:1 (5)
Class II	11	6	4:2 (2)
Class II MOD	14	8	3:3 (1)
Class III	6	3	3:3 (1)
Class IV	11	6	2:4 (0.5)
Class V	8	4	5:1 (5)
• *Non-carious cervical lesion (abrasion/abfraction)*			1:5 (0.2)

63. Preventive resin restorations are indicated in?

A. Non-cavitated pit and fissures
B. Deep caries to prevent progress of lesion
C. Cavitated fissures which require restoration
D. Wide cavity to prevent cusp fracture

FISSUROTOMY

64. During cavity preparation, isolating the proximal enamel is known as:

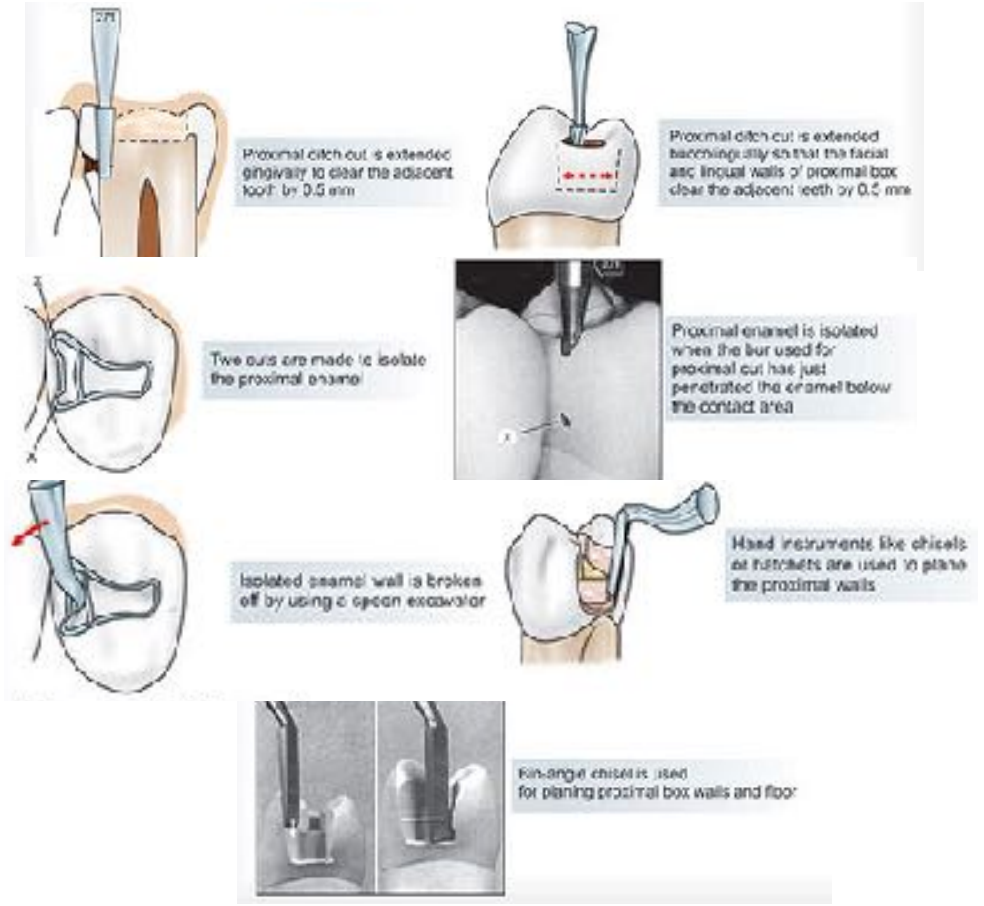
A. Pitching

B. Ditching

C. Undercutting

D. Overcutting

Proximal ditch cut



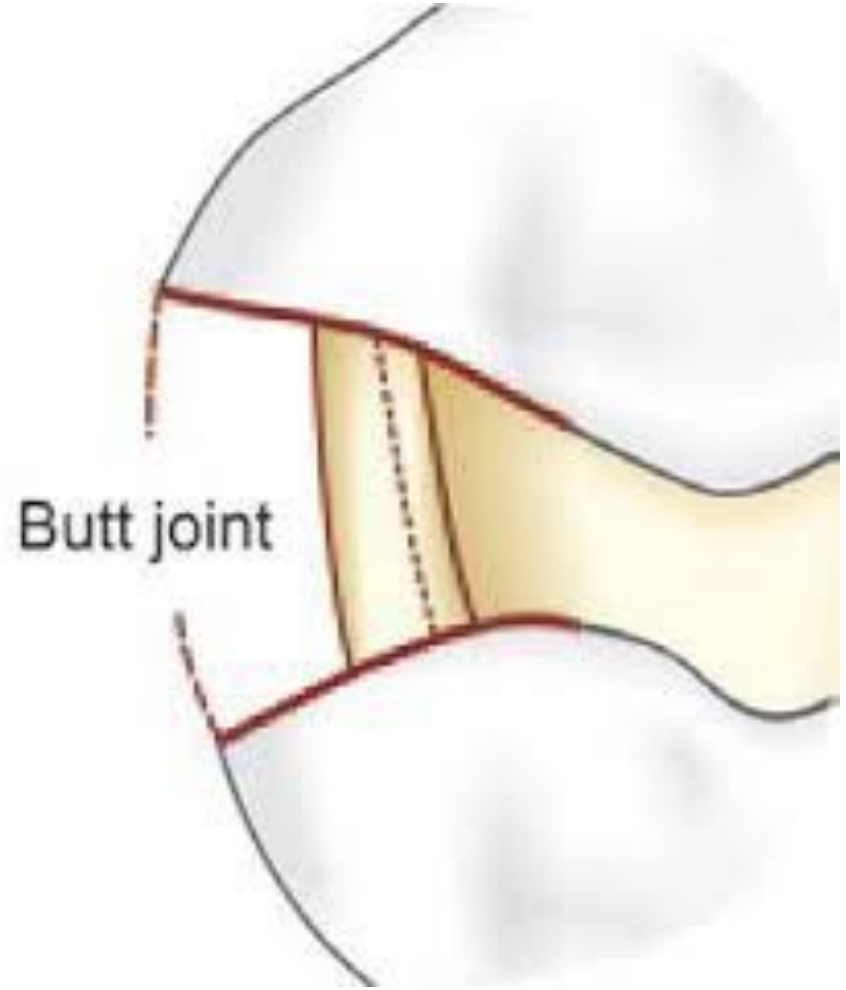
65. A buttjoint is a

A. 90 degree joint

B.120 degree joint

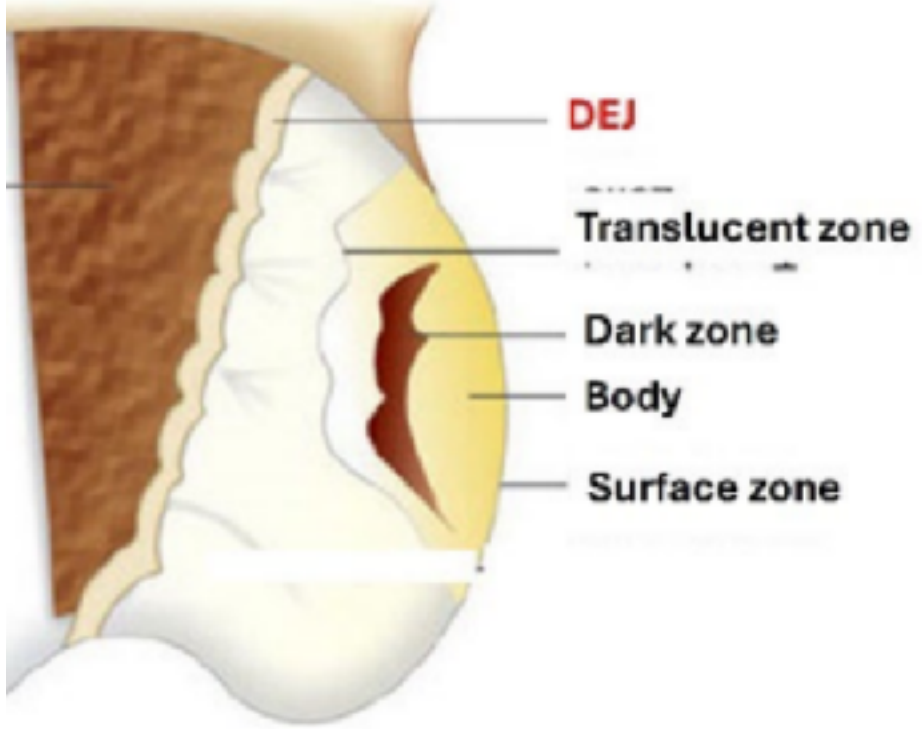
C.70 degree joint

D.180 degree joint



66. Incipient caries consists of opaque, chalky white areas that appear when the tooth surface is dried, this is referred as

A. White spot
B. Hot spot
C. Translucent zone
D. Body of lesion



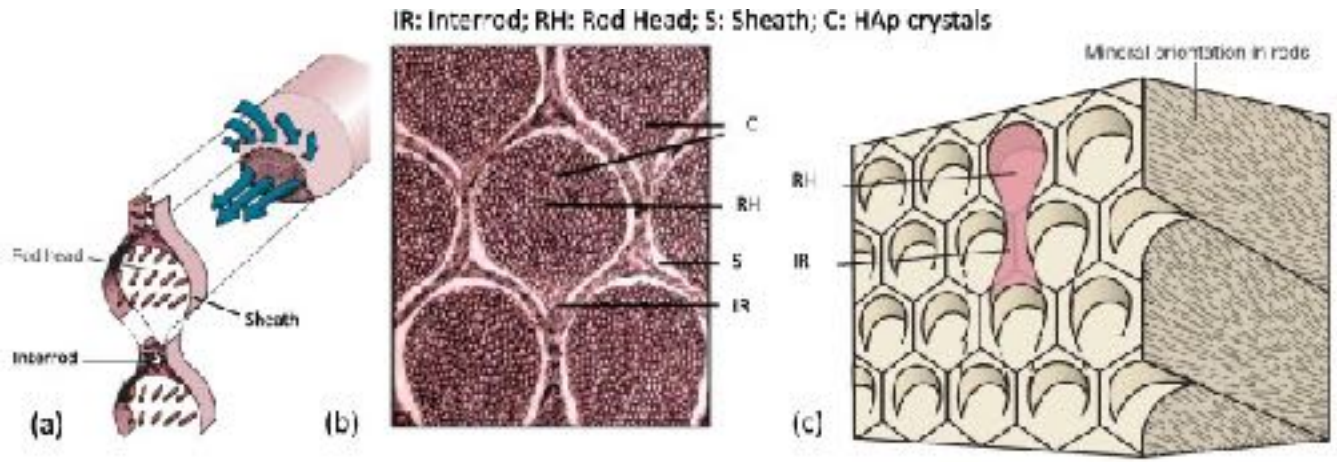
ICDAS

International Caries Detection & Assessment System

ICDAS II

Score 0	Score 1	Score 2	Score 3	Score 4	Score 5	Score 6
No visual signs of carious lesions or any enamel defect	First visible changes in the enamel. Visible only after drying with air. Changes in coloration confined to areas of pits	Change in visible enamel even in the presence of moisture. More extensive and not restricted to pits.	Destruction located in enamel without visible dentin, discontinuities of enamel surface	Dark shadow on the underlying dentin, with or without localized destruction of enamel	Clear cavity with visible dentin; cavity that involves less than half the dental surface	Extensive cavity evident in dentin; cavity deep and wide, involves more than half of the tooth

Loss of Interprismatic substance



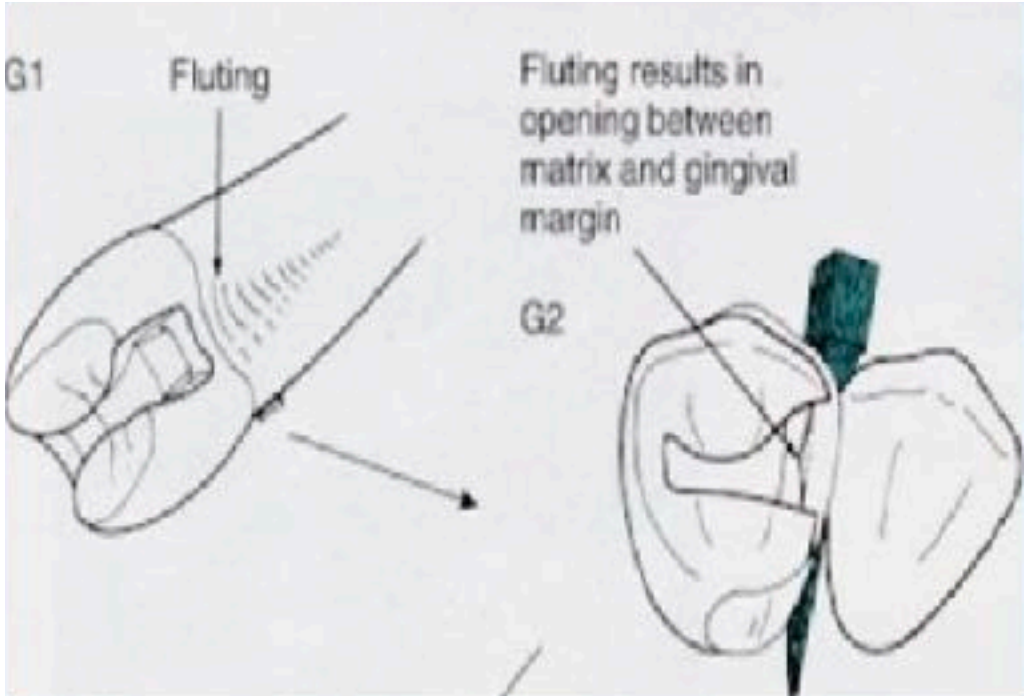
67. Matrix band is difficult to place on _____ tooth

A.MO 25

B.DO 24

C.DO 37

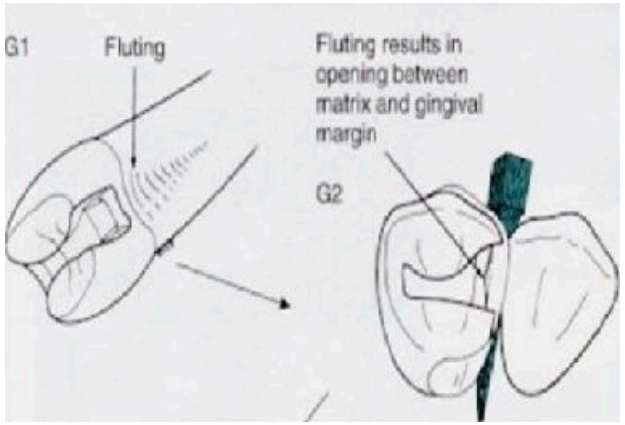
D.MO 24

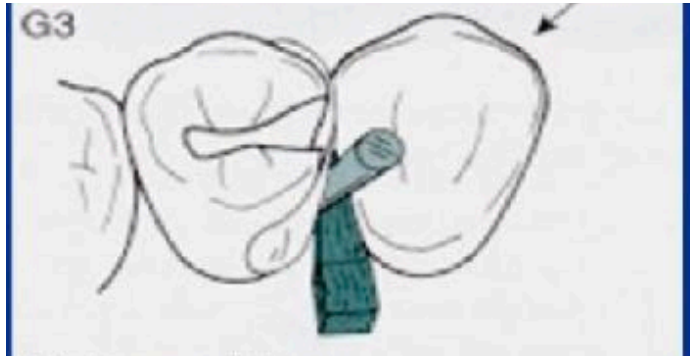


TYPES OF WEDGING

4. WEDGE WEDGING

-a 2nd wedge is inserted from lingual or facial embrasure, between the 1st wedge & the band





INDICATION:
proximal concavity, gingival to the contact (mesial of maxillary 1st premolars)

68. As caries progresses from the enamel to dentin and toward the pulp interproximally, the radiographic appearance resembles which of the following?

A. V shape in enamel with apex toward the proximal surface and V shape in dentin with apex toward the pulp, resulting in a diamond appearance
B. V shape in enamel with apex toward the the DEJ and V shape in dentin with apex toward the DEJ, resulting in a bowtie appearance
C. V shape in the enamel with the apex toward the DEJ and V shape in dentin with apex toward the pulp
D. V shape in enamel with apex toward the proximal surface and V shape in dentin with apex toward the DEJ

69. For sandblasting ceramic veneer, the particle size of alumina used is

A. 25 micrometer
B. 50 micrometer
C. 75 micrometer
D. 100 micrometer

Air abrasion/ sandblasting

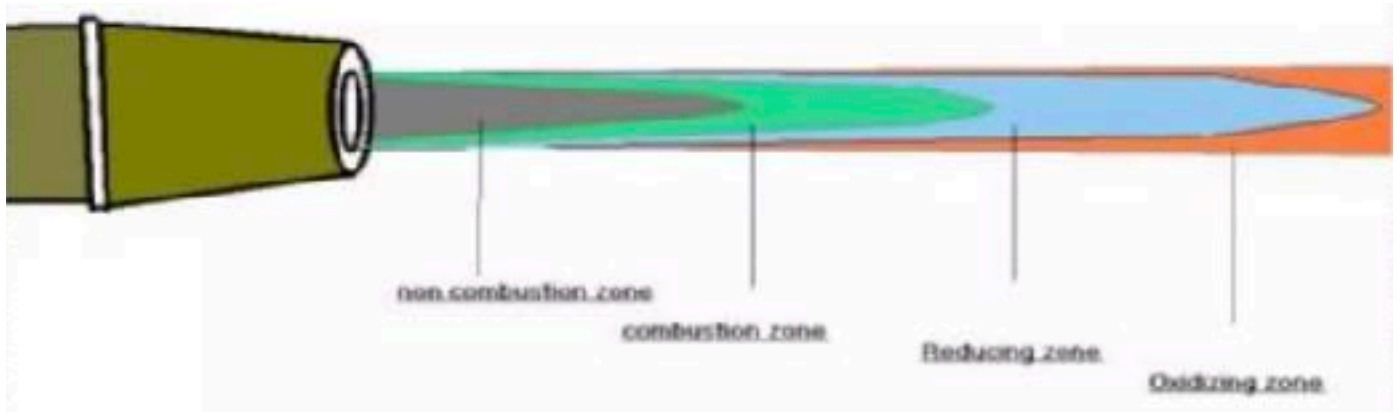
Aluminum oxide powder

Air polish/Air flow

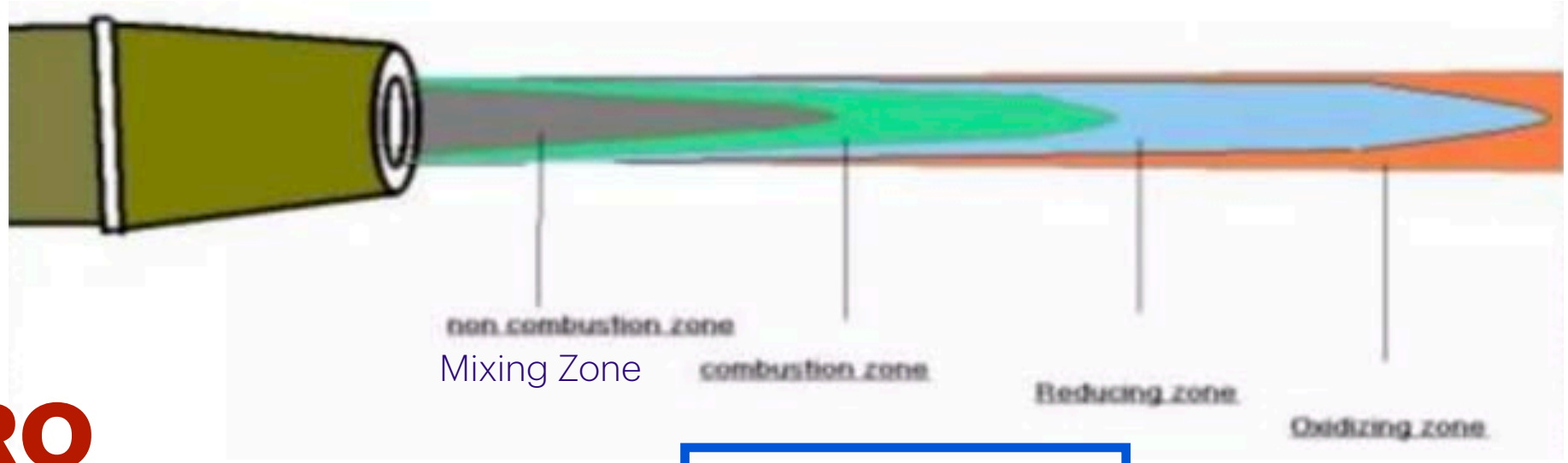
-Sodium bicarbonate powder
-Glycine powder
-zirconium silicate powder

70. Which of the following zones is used for melting alloy during casting?

A. Reducing flame
B. Oxidizing flame
C. Green flame
D. All of the above



ZONES OF TORCH FLAME (GAS AIR BLOW PIPE)



MiCRO

Colorless Zone(Directly from the nozzle)	Combustion Zone(Green Colour)	Reducing Zone(Blue color)	Oxidizing Zone
•Air and gas are mixed before combustion . •No heat is present	•Gas and air are partially burned. •Oxidizing and should be kept away	•Hottest part of flame •Most effecting zone and should be always in contact with alloy while melting	•Combustion occurs with the oxygen in the air

71. Comparing a composite restoration and an amalgam restoration in a Class I cavity, composite is inferior to amalgam with respect to _____

A.Creep
B.Fracture
C.Occlusal wear
D.Marginal breakdown

72. The curing of direct filling composite resin is inhibited by:

A. Calcium hydroxide
B. Fillers
C. Moisture
D. Oxygen

OXYGEN INHIBITION LAYER

When composite is light cured, oxygen in air causes interference in polymerization forming **oxygen inhibition layer.**

This layer is ***sticky, resin-rich uncured layer left on the surface.***

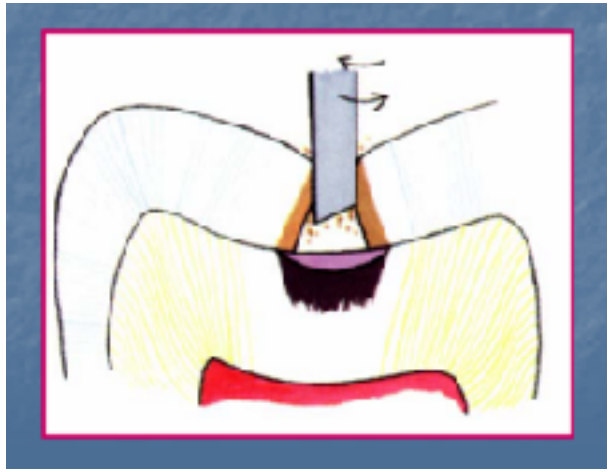
This allows resin to interface & form interdiffusion zone (ALLOWS INCREMENTS)

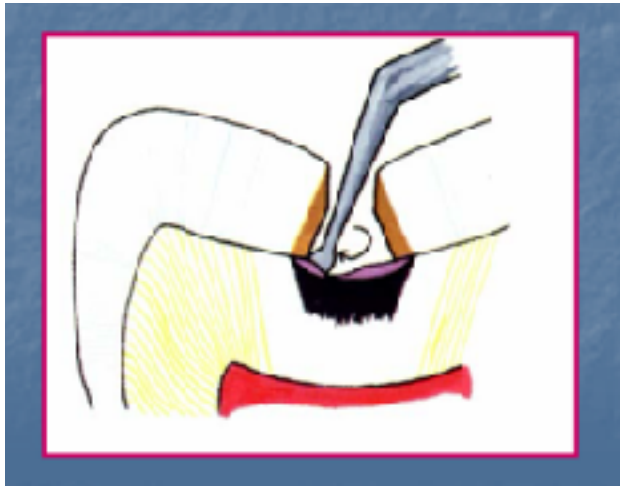
After the final layer, remove this layer for a **smooth surface finish**

*__Glycerine gel__ before final curing also helps eliminate this layer

73. During ART procedure, which instrument is used to remove carious dentin?

A. Ultrasonic scaler
B. High-speed handpiece
C. Air abrasion unit
D. Excavator



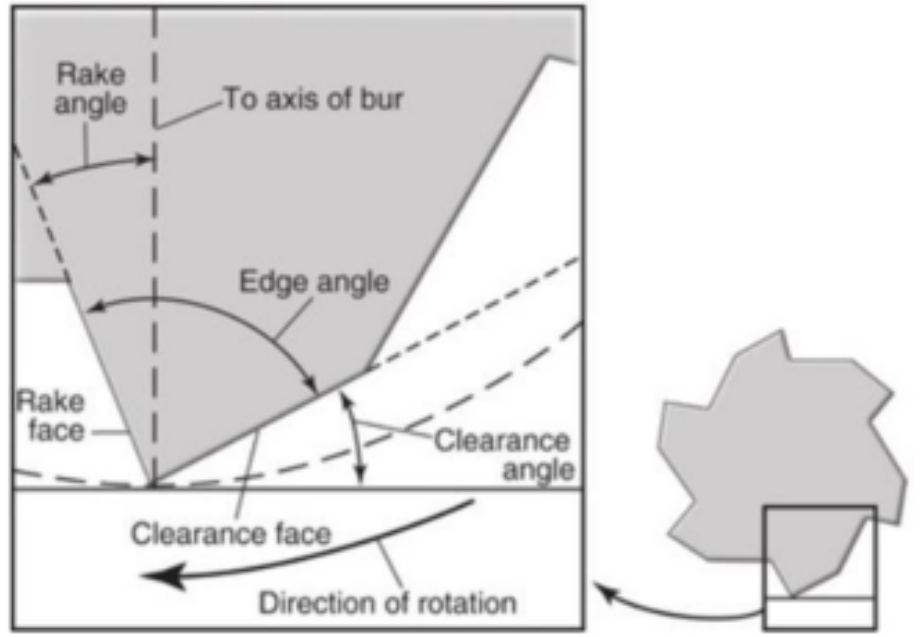


74. The _____ angle is the angle formed between the _____ face and the _____ face

A. Edge, rake, clearance

B. Rake, edge, clearance

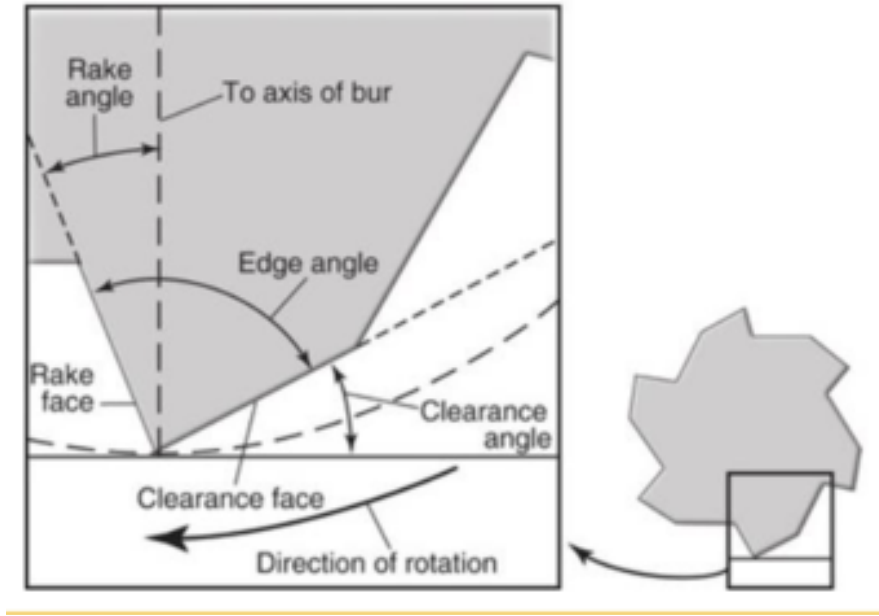
C. Clearance, rake, edge



2 sides of bur blade:

***rake face:** toward direction of cutting, makes contact with the tooth surface & face

***clearance face:** side that faces away from the direction of bur rotation



3 ANGLES OF BUR BLADE:

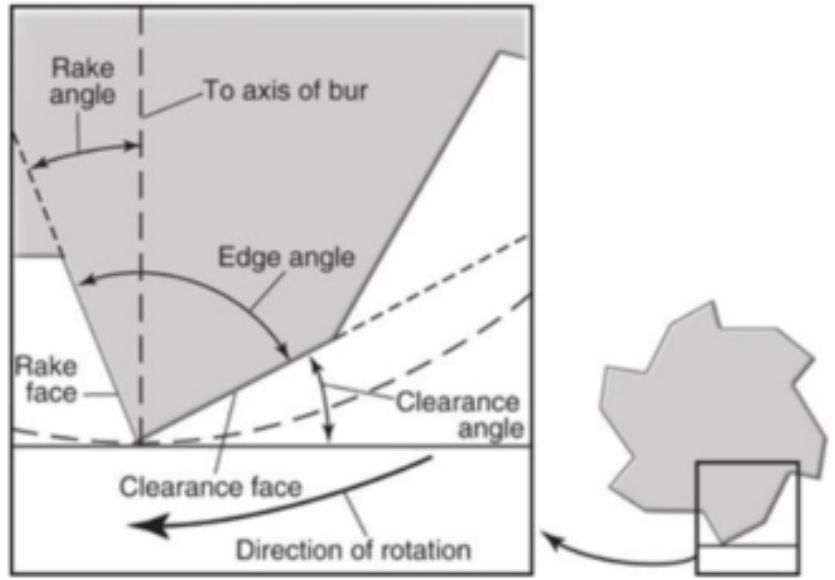
***rake angle:** between the line connecting the edge of the blade to the axis of bur & rake face. This maybe positive or negative

***edge angle:** between the rake face and clearance face

***clearance angle:** between the clearance face and a tangent to the path of rotation

Negative rake angle: when the rake face is ahead of the radius (axis of bur), for extremely hard and brittle materials

Positive rake angle: for soft materials like acrylic



75. The component of composite resins which provides working time during restoration build up and prolong storage life by preventing polymerization when not in use

A. UV absorber BENZOPHENE: Provides color stability

B. Optical modifiers Metal oxides: provides opacity or translucency

C. Fillers Determines strength

D. Hydroquinone *butylated hydroxytoluene (BHT): INHIBITOR

E. Camphorquinone initiator for light curing resin

76. What is the blade width in mm of an instrument with a formula 10-8-15

A.10mm

B.1.0mm

C.8.0mm

D.0.8mm

HAND CUTTING INSTRUMENTS

4-DIGIT FORMULA

1	W: width of blade in tenths of mm
2	A: angle formed by cutting edge and long axis of shaft *measured in degrees centigrade (always exceeds 50); 360 degrees =680 Fahrenheit 100 degrees centigrade
3	L: length of blade in mm *2nd digit in 3 numbered instrument
4	A: angle formed by blade with long axis of shaft

77. What is the blade angle of an instrument with a formula 10-8-15

A.23.8 degrees
B.36 degrees
C.54 degrees
D.15mm

78. A high copper amalgam is preferred over a low copper amalgam due to the undesirable clinical properties that present in which phase of the reaction?

A. Gamma
B. Gamma-1
C. Gamma-2
D. Gamma-3

Phases of Amalgam Reaction

- **Gamma Phase** Ag_3Sn (γ)
  - *Silver + Tin alloy*
  - Strongest and Corrodes least
  - 30% of amalgam
- **Gamma-One Phase** Ag_2Hg_3 (γ1)
  - *Silver + Mercury*
  - Second Strongest phase
  - 60% of amalgam – most numerous
- **Gamma-Two Phase** Sn_8Hg (γ2)
  - *Tin + Mercury*
  - Weakest and softest; Corrodes most
  - Prone to microporosities

Low Copper Amalgam

<6% Copper or 2-5%

- Has gamma-two phase

High Copper Amalgam

>9%-30% copper or 13%

- Eliminates **GAMMA-TWO PHASE**

Cu_6Sn_5

79. Final polishing of dental amalgam to get a smooth surface is achieved by

A. Burnishing

B. Use of tin oxide

C. Carving with finer instruments

D. Use of iron oxide

POLISHING

Restoration	What	When
Composite	Aluminum oxide, silicon carbide, silica grit, diamond grit	Immediately after restoration (10-15mins)
Gold	Iron oxide	After final condensation &contouring
Amalgam	Tin oxide	After 24 hrs

*to avoid disturbing crystallization of amalgam
*to avoid cracking/fracturing the incompletely cryztallized restoration

80. If the setting reaction of Glass ionomer is acid-base reaction, what is the setting reaction of amalgam?

A. Crystallization

B. Polymerization

C. Oxidation-reduction

D. Acid-base reaction

Amalgam & Gold

81. The coupling agent used in composite resin

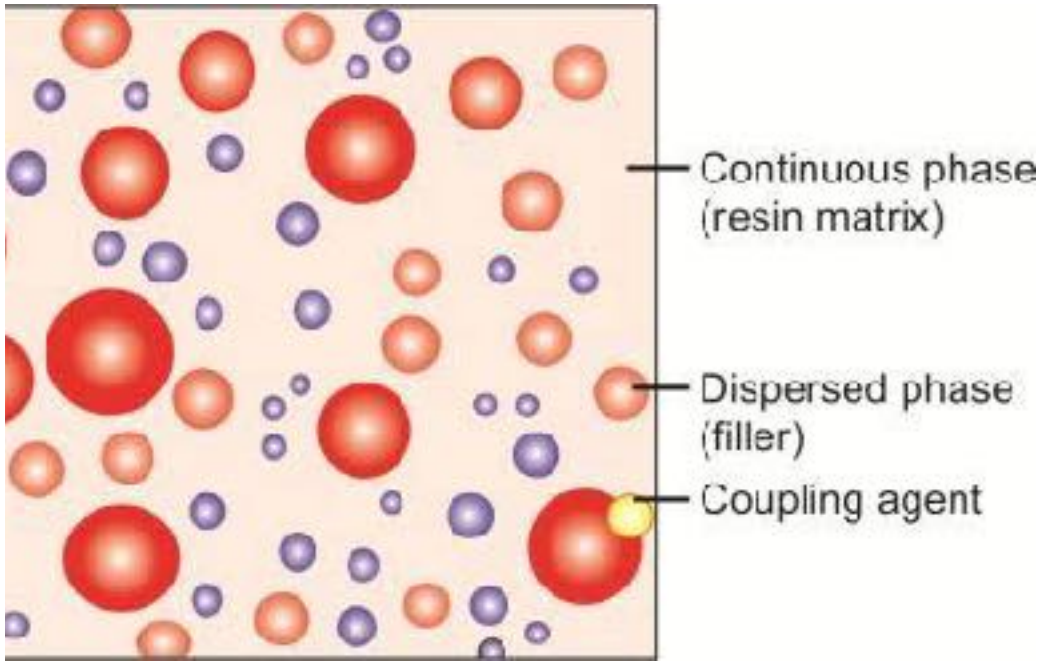
A. Colloidal silica -fillers

B. Urethane dimethacrylate -matrix

C. Organo silane

D. Diketone -initiator for light curing composites

COMPOSITE (ADA spec 27)



Matrix/Monomers:

Bis-GMA
UDMA
*TegDMA

Coupling Agent:

Promotes adhesion (covalent bonds) between fillers & resin matrix

Filler: determines the strength of the composite (70% by wt)

Crystalline silica (quartz), Lithium aluminum silicate	-radiolucent
Barium glass, zinc, boron, zirconium, and yttrium	-radiopaque
Colloidal silica	

82. Discolored areas seen through the enamel in the teeth that have amalgam restorations are known as

A. Amalgam tattoos
B. Amalgam overhangs
C. Amalgam blues
D. Amalgam ditchings

AMALGAM BLUES

AMALGAM TATTOO

AMALGAM DITCHINGS

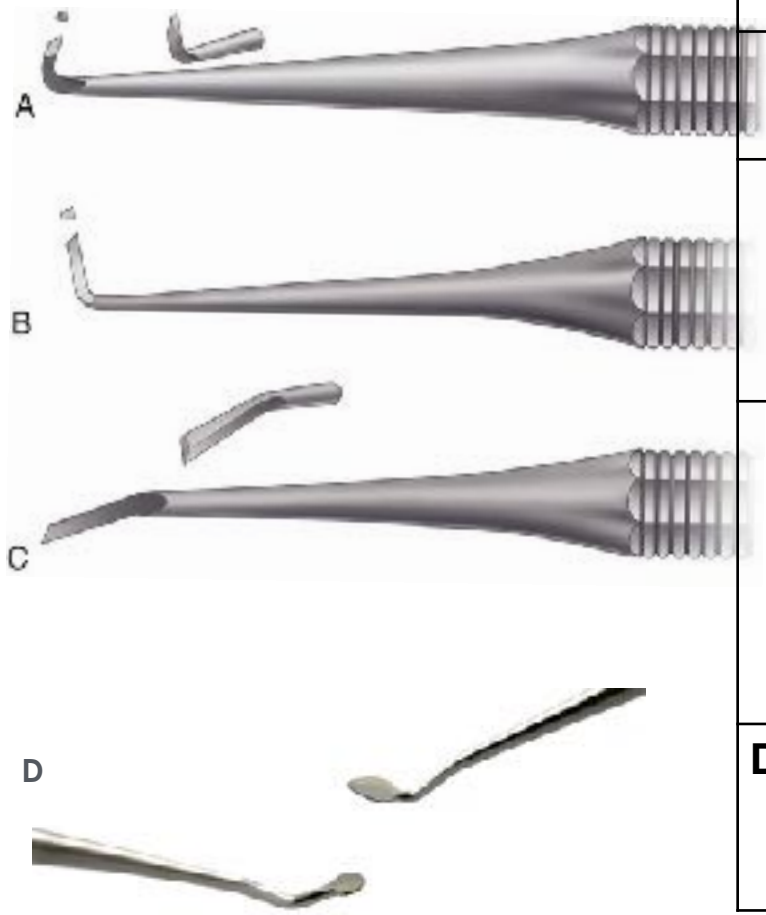
83. To avoid pulpal irritation below a resistant metallic restoration, the minimum dentin thickness which should remain is:

A. 0.5mm
B. 1.5mm
C. 1.0mm
D. 2.0mm

84. An instrument used for planing walls and sharpening line angles especially in Class III and Class V direct gold restorations

A. Enamel hatchet
B. Finishing bur
C. Hoe
D. Gingival margin trimmer

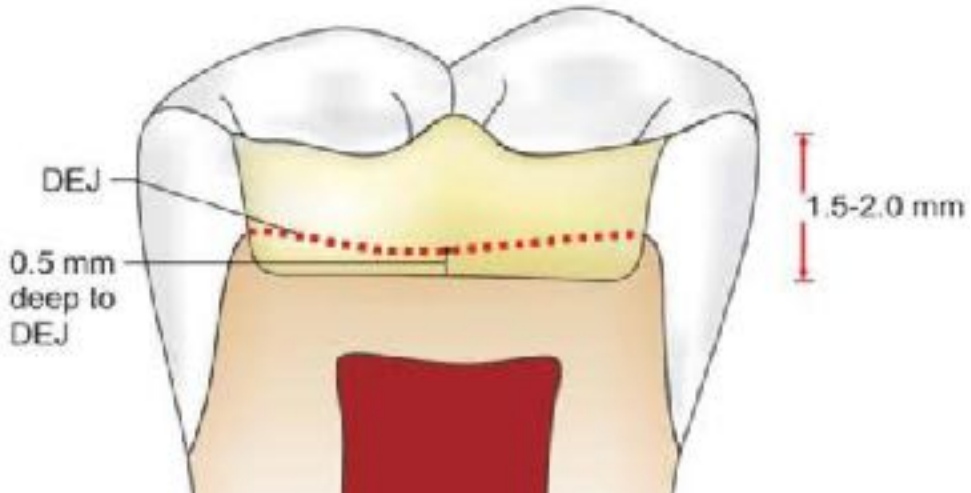
- **Examples of Instruments and their Use**
  - **<u>EXCAVATORS</u>** – used mainly to remove caries and refine internal walls of preparation

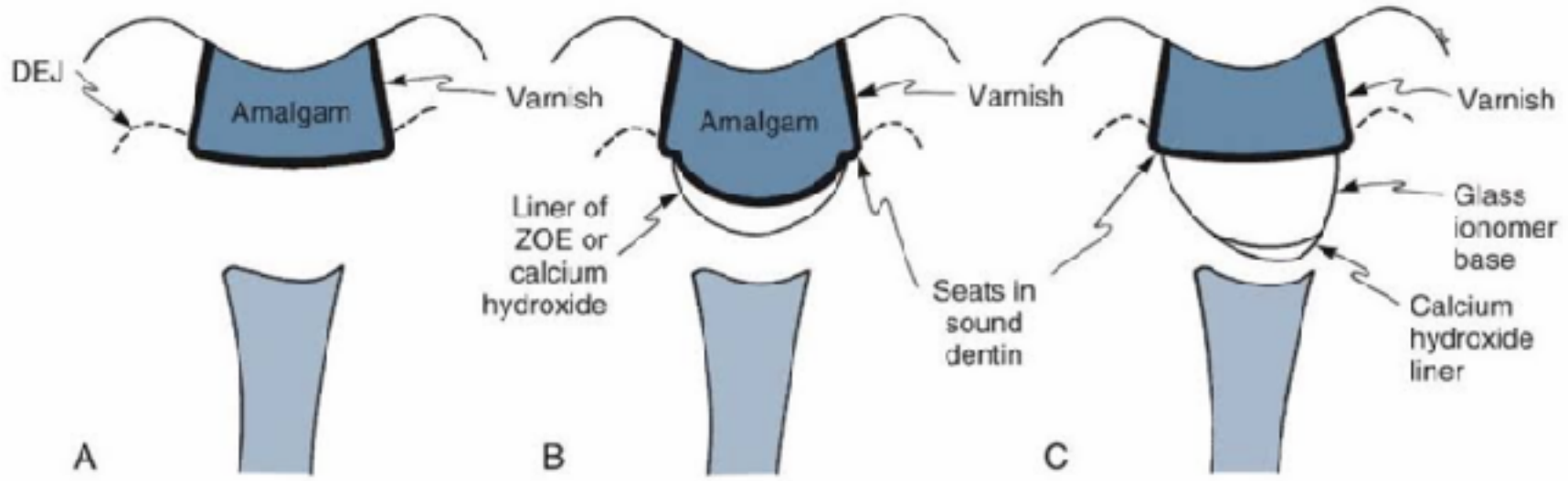


Types of Excavators	
Ordinary Hatchets	➢ Create retentive features ➢ Sharpening internal line angles
Hoe	➢ For Planing walls ➢ Sharpening line angles ➢ Especially in Class III and Class V direct gold restorations ***Note Hoe has 3 letters (Class III)*
Angle Former	➢ Sharpening line angles ➢ Create retentive features ➢ Enamel margin beveling ➢ Combination of Gingival Marginal Trimmer and Chisel ➢ Mono angled only (W/ blade angle of more than 125 degrees)
Discoid and Cleoid (spoons)	➢ Caries removal, Amalgam carving ➢ Circular (discoid) and clawlike (cleoid) end ➢ Can be Bin-angled or Triple-angled

85. What is the ideal depth of an amalgam preparation and dentin remaining for adequate pulpal insulation?

A. 1.0mm into dentin, 2.0mm dentin remaining
B. 2.0mm into dentin, 1.0mm dentin remaining
C. 0.5mm into dentin, 2.0mm dentin remaining
D. 2.0mm into dentin, 0.5mm dentin remaining





TYPES OF CAVITY PREPARATION (DEPTH)		
Class A	**Class B**	**Class C**
Shallow	Moderately deep	Deep excavations
2mm or more RDT	0.5-2mm RDT	<0.5mm RDT
*2 coats varnish/sealer *amalgam	*ZOE liner or GI liner *varnish/sealer *amalgam	*CaOH liner *GI base *varnish/sealer *amalgam *(indirect pulp capping)*

86. When restoring with a darker shade of composite, keep in mind the following:

A. UV light is better than visible light
B. You must keep the light 2mm away or more
C. You should cure for a longer than normal
D. Darker shades have less chemical bonding

87. A gold onlay you placed last week fails. Which of the following reasons is most likely responsible for failure?

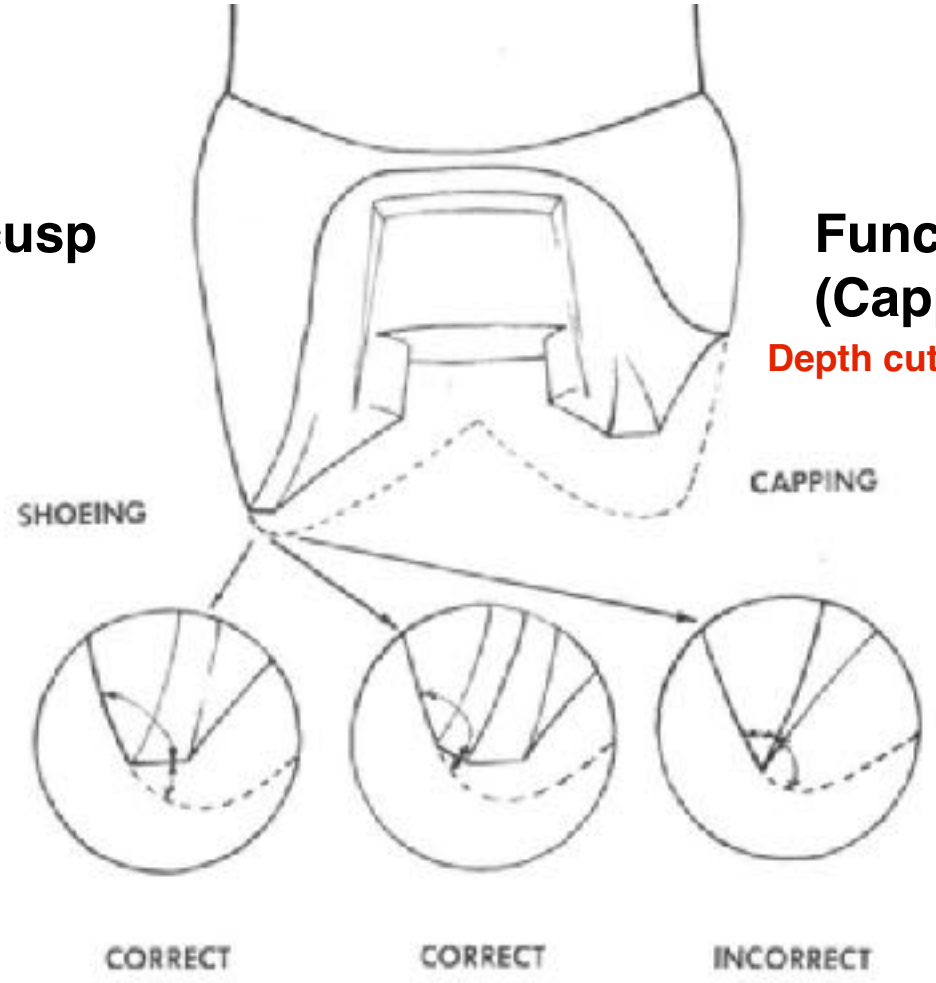
A. You “capped” the functional cusp
B. You “shoed” the functional cusp NEVER INDICATED
C. You “capped” the nonfunctional cusp
D. You “shoed” the nonfunctional cusp

**capping is always preferred over shoeing, unless case demands minimal display of gold (facial of premolars & molars)*

Non-functional cusp (Shoeing)

Functional cusp (Capping)

Depth cuts: minimum of 2.0mm



Non-Functional cusp capping

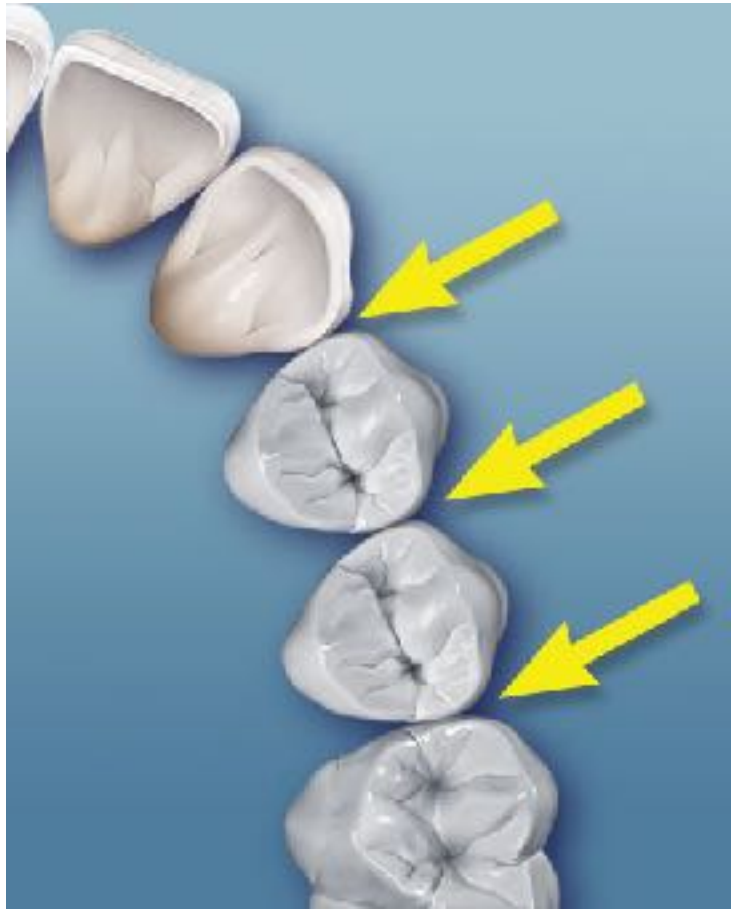
Depth cuts: minimum of 1.5mm

88. Chisels are used primarily to cut dentin. Hatchets are primarily used to cut enamel

A. Both statements are true
B. Both statements are false
C. The 1st statement is true, the 2nd statement is false
D. The 1st statement is false, the 2nd statement is true

89. When restoring the embrasures of the posterior teeth, the contact should be formed slightly buccal from center. This will create a wider facial embrasure

A. Both statements are true
B. Both statements are false
C. The 1st statement is true, the 2nd statement is false
D. The 1st statement is false, the 2nd statement is true



90. Glass ionomer cements generally contain fluoro-amino-silica powder. Fluoro-amino-silica powder is the portion responsible for one of the major advantage of glass ionomer

A. Both statements are true
B. Both statements are false
C. The 1st statement is true, the 2nd statement is false
D. The 1st statement is false, the 2nd statement is true

91. IRM (intermediate restorative material) will interfere with subsequent placement of a resin filling. This is because IRM is a form of zinc oxide eugenol

A. Both the statement and the reason are correct and related
B. Both the statement and the reason are correct, but not related
C. The statement is correct, but the reason is not
D. The statement is not correct, but the reason is correct
E. Neither the statement nor the reason is correct

92. The wall absent in class V lesion

A. Axial wall
B. Mesial
C. Distal
D. Pulpal wall

93. What is sandwich technique

A. Composite is sandwich between cement base and amalgam
B. Polycarboxylate cement replace dentin and composite replace enamel
C. Composite replace dentin and GIC replace enamel
D. RMGI is placed as a liner beneath composite restoration where RMGI will act as dentin and composite as enamel

Closed Sandwich

GIC is completely covered by composite resin

Open Sandwich

GIC remains exposed to oral cavity

94. What is the powder used in picking up amalgam/ mercury spillage?

A. Aluminum oxide
B. Calcium hydroxide
C. Sulfur
D. Eugenol

95. A dental clinician accidentally dropped a capsule of dental amalgam on a tile floor in the dental clinic. Which of the following is the most appropriate initial action to gather the spilled mercury droplets?

A. Use a high-volume dental evacuator or household vacuum to suction the beads
B. Use a broom and dustpan to sweep the beads into a biohazard bag
C. Use a rubber squeegee or stiff cardboard to gently push the beads together
D. Wash the area immediately with hot water and heavy-duty detergent

96. Fractured incisors and large Class IV cavities is resorted using this type of matrix

A. Celluloid strips
B. Banding strips
C. Clear plastic crown forms
D. Copper bands

97. When finishing the lingual surface of a composite restoration in an anterior tooth, the most effective abrasive tool would be a:

A. Finishing disk
B. Finishing cup
C. Flame-shaped finishing bur
D. Egg/football-shaped finishing bur

98. During preparation for a porcelain laminate veneer on a maxillary central incisor, the dentist follows a three-zone facial reduction technique. Which of the following correctly matches the recommended facial reduction depth for each third of the tooth?

A. Incisal: 0.3-0.5mm, middle: 0.5-0.75mm, gingival: 0.75-1.0mm
B. Incisal: 0.75-1.0mm, middle: 0.5-0.75mm, gingival: 0.3-0.5mm
C. Incisal: 1.0-1.5mm, middle: 0.75mm-1.0mm, gingival: 0.5-0.75mm
D. Incisal: 0.5-0.75mm, middle: 0.3-0.5mm, gingival: 0.75-1.0mm

99. A dentist uses an adhesive system that requires separate application of phosphoric acid etchant, primer, and bonding resin. This system is characterized by 3 distinct clinical steps. Which generation of dentin bonding system does this technique represent?

A. Fourth generation
B. Fifth generation
C. Sixth generation
D. Seventh generation

100. Resin modified glass ionomer is designed to

A. Provide rapid setting to minimize cement acidity allowing early finishing
B. Reduce the rate of fluoride release
C. Reduce the cost of te product
D. Change the translucency of the products

Prayer for Examination Time

God of wisdom: guide me as
I choose questions and in the way
I answer them.

God of peace: keep me calm so that
I will do the best that I can.

God of understanding: help me put
all of this into perspective.

I ask these prayers in the name
of Jesus, friend of students.
Amen.

1 month to go....

GOOD LUCK & GOD BLESS, DOCTORS!!